

Continuous Computation

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CHAPTER I
Measure Theory

1.1 Measure

Measure and integration theory develops the necessary tool for solving PDEs and stochastic processes.

σ -Algebra

Motivation In physics, the task of finding “volume” is ubiquitous. For instance in computing mass, charge, the integral would resemble:

$$\int (\cdot) dV \quad (1)$$

In some sense, this dV is a function $dV : E \rightarrow \mathbb{R}$ which computes the volume of $E \subset \mathbb{R}^n$, a tiny control volume we are interested in. It would be very natural to ask for this function to satisfy:

Definition 1.1 (Measure)

Suppose X is a set. Let $\mu : A \rightarrow \mathbb{R}$ where $A \in P(X)$ such that:

-

$$\mu(\emptyset) = 0 \quad (2)$$

- **(countable additivity)** $E_1, E_2, \dots$ is a countable infinite sequence of disjoint, then:

$$\mu(\cup_j E_j) = \sum_j \mu(E_j) \quad (3)$$

Then we call μ a measure.

The question would be, when can we find such a function. Trivially we can obtain such property by sending every A to 0. However, for more practical purposes of modeling the world, we would like nicer properties. In particular, let's consider this completely reasonable demand on $\mathbb{R}^n$.

Exercise 1.1 (Open Set as a Topology)

1. Let $\mathcal{B}$ be the family of all open intervals in $\mathbb{R}$, i.e. subsets of the form

$$(a, b) := \{c \in \mathbb{R} \mid a < c < b\} \quad (4)$$

for all $a, b \in \mathbb{R}$ with $a < b$. Let $\mathcal{T}(\mathcal{B})$ be the family of all arbitrary unions of elements in $\mathcal{B}$, i.e.,

$$\mathcal{T}(\mathcal{B}) := \left\{ \bigcup_{B \in \mathcal{B}'} B \mid \mathcal{B}' \subseteq \mathcal{B} \right\} \quad (5)$$

Prove that $\mathcal{T}(\mathcal{B})$ is a topology on $\mathbb{R}$. (This “ $\mathcal{B}$ ” stands for a “base” = “basis” of a topology.)

2. For $\mathcal{B}$ and $\mathcal{T}(\mathcal{B})$ as above and any $U \in \mathcal{T}(\mathcal{B})$ (i.e., any open set), prove that there exists a countable family $\mathcal{B}'' \subseteq \mathcal{B}$ (of open intervals) such that $U = \bigcup_{B \in \mathcal{B}''} B$. (Hint: it might help to prove first that the union of any family of open intervals containing a given $x \in \mathbb{R}$ is a generalized open interval in the sense that it is of the form (a, b) , $(-\infty, a)$ or (a, ∞) for some $a, b \in \mathbb{R}$. Or do this any other way.)
3. For $\mathcal{T}(\mathcal{B})$ as above, prove that $\mathcal{T}(\mathcal{B}) \subseteq \mathcal{B}_{\mathbb{R}}$, the Borel σ -algebra on $\mathbb{R}$. (This “ $\mathcal{B}$ ”, confusingly, stands for “Borel”.)
4. Use the above to give a full proof of Proposition 1.2 on p. 22 in [F].
5. Also deduce that $\mathcal{M}(\mathcal{B}) = \mathcal{M}(\mathcal{T}(\mathcal{B})) = \mathcal{B}_{\mathbb{R}}$.

Solution to Exercise 1.1

1. We prove that $\mathcal{T}(\mathcal{B})$ is a topology on $\mathbb{R}$.

• **Property 1:** $\emptyset, \mathbb{R} \in \mathcal{T}(\mathcal{B})$

- $\emptyset = \cup_{B \in \emptyset} B$ (empty union), so $\emptyset \in \mathcal{T}(\mathcal{B})$
- $\mathbb{R} = (-\infty, \infty)$ is a generalized open interval, hence $\mathbb{R} \in \mathcal{T}(\mathcal{B})$. More explicitly, we can cover it by $\cup_{a=0}^{\infty} \{(a, a+1), (-a-1, -a)\}$

• **Property 2:** Closed under arbitrary unions

- Let $\mathcal{U} \subseteq \mathcal{T}(\mathcal{B})$ be an arbitrary collection of sets in $\mathcal{T}(\mathcal{B})$. For each $U \in \mathcal{U}$, we have $U = \cup_{B \in \mathcal{B}_U} B$ for some $\mathcal{B}_U \subseteq \mathcal{B}$. Then $\cup_{U \in \mathcal{U}} U = \cup_{U \in \mathcal{U}} (\cup_{B \in \mathcal{B}_U} B) = \cup_{B \in \cup_{U \in \mathcal{U}} \mathcal{B}_U} B$. Since $\cup_{U \in \mathcal{U}} \mathcal{B}_U \subseteq \mathcal{B}$, we have $\cup_{U \in \mathcal{U}} U \in \mathcal{T}(\mathcal{B})$

• **Property 3:** Closed under finite intersections

- By induction on the number n of sets.
- **Base case** ($n = 1$): Trivial, $U \in \mathcal{T}(\mathcal{B})$.
- **Inductive step:** Assume $U_1 \cap \cdots \cap U_n \in \mathcal{T}(\mathcal{B})$ for any n sets in $\mathcal{T}(\mathcal{B})$. For $n + 1$ sets, write $(U_1 \cap \cdots \cap U_n) \cap U_{n+1}$. By induction hypothesis, $U_1 \cap \cdots \cap U_n = \cup_{i \in I} (a_i, b_i)$ for some countable collection of open intervals. Then

$$(U_1 \cap \cdots \cap U_n) \cap U_{n+1} = (\cup_{i \in I} (a_i, b_i)) \cap U_{n+1} = \cup_{i \in I} ((a_i, b_i) \cap U_{n+1}) \quad (6)$$

Each $(a_i, b_i) \cap U_{n+1}$ is an intersection of an open interval with a union of open intervals, which is a finite union of open intervals (or empty). Hence $(U_1 \cap \cdots \cap U_{n+1}) \in \mathcal{T}(\mathcal{B})$.

Therefore $\mathcal{T}(\mathcal{B})$ is a topology on $\mathbb{R}$.

2. Suppose $\mathcal{B}' \subseteq \mathcal{B}$ and

$$U = \bigcup_{B \in \mathcal{B}'} B \quad (7)$$

where $\mathcal{B}'$ can be uncountable. We would like to show $\exists \mathcal{B}''$ that is countable such that

$$U = \bigcup_{B \in \mathcal{B}''} B \quad (8)$$

We now show a specific way to construct $\mathcal{B}''$ from $\mathcal{B}'$. Define:

$$\mathcal{B}_x = \{(a, b) \in \mathcal{B}' : x \in (a, b)\} \quad (9)$$

Clearly $\mathcal{B}_x \subseteq \mathcal{B}'$. Now proceed with the following procedure (an “infinite algorithm”) on the construction of $\mathcal{B}''$

Algorithm 1.1**Input:** A potentially uncountable set of open intervals $\mathcal{B}'$ such that $\cup \mathcal{B}' = U$ **Output:** A countable set of open intervals $\mathcal{B}''$ such that $\cup \mathcal{B}'' = U$

1. $\mathcal{B}'' = \emptyset$
2. **While** $\exists x \in U, x \notin \cup \mathcal{B}''$:
 - (a) Compute $(c, d) = \bigcup_{B \in \mathcal{B}_x} B$ where c, d is in extended real number system.
 - (b) **If** $c, d \in \mathbb{R}$:
 - i. $\mathcal{B}'_x = \{(c, d)\}$
 - (c) **Else if** $c \in \mathbb{R}, d = \infty$:
 - i. $\mathcal{B}'_x \leftarrow \{(c, [c])\} \cup \{[c], [c] + j\}_{j=1}^{\infty}$
 - (d) **Else** Other cases follows as (3)
 - (e) $\mathcal{B}'' = \mathcal{B}'' \cup \mathcal{B}'_x$
3. **Return** $\mathcal{B}''$

where $[c]$ picks the closes integer towards the right of the real axis. Note we implicitly invoked the claim below in computing (c, d) :

- **Cliam 1** The union of arbitrary family of open intervals containing $x \in \mathbb{R}$ is a *generalized open interval*

$$\bigcup_{B \in \mathcal{B}_x} B \in \{(c, d) : c, d \in \mathbb{R} \cup \{\infty, -\infty\}\} \quad (10)$$

- I claim the union is just (c, d) where $c = \inf\{a : (a, b) \in \mathcal{B}_x\}$ and $d = \sup\{b : (a, b) \in \mathcal{B}_x\}$ where we allow ∞ and $-\infty$. Clearly (c, d) covers $\cup \mathcal{B}_x$. It suffice to show $\cup \mathcal{B}_x$ covers (c, d) . Suppose $\exists y \in (c, d)$ and $y \notin \cup \mathcal{B}_x$. This means $\cup \mathcal{B}_x$ has a hole and there must exists a pair of open interval in $\mathcal{B}_x$ that is disjoint, contradicting that all open intervals in $\mathcal{B}_x$ contains x (in that case no pair should be disjoint).

Now what's left to show is (1) Only countable number of $\mathcal{B}'_x$ is needed. (2) each $\mathcal{B}'_x$ is countable. The later is obvious by construction. The former can be argued by countability of $\mathbb{Q}$. Pick a $q \in \mathbb{Q}$ in one of the open interval of $\mathcal{B}'_x$, this is always possible by density of rationals. Suppose uncountable $\mathcal{B}'_x$ is needed to cover $\cup \mathcal{B}''$ then $U = \cup \mathcal{B}''$ would contain uncountable number of rationals. This finishes the proof.

-
3. By definition, $\mathcal{B}_{\mathbb{R}}$ is the smallest σ -algebra containing $\mathcal{T}(\mathcal{B})$. Therefore the inclusion holds by construction.
-

4. We only need to show (b), (c), and (e) since (a) and (d) are generalized open intervals and are actually shown by (5).

Case (b): Closed intervals $\mathcal{E}_2 = \{[a, b] : a < b\}$

- $\mathcal{M}(\mathcal{E}_2) \subseteq \mathcal{B}_{\mathbb{R}}$: Note that $[a, b] = \cap_{j=1}^{\infty} (a - 1/j, b + 1/j)$ is a countable intersection of open intervals, so $\mathcal{E}_2 \subset \mathcal{B}_{\mathbb{R}}$.
- $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_2)$: $(a, b) = \cup_{j=1}^{\infty} [a + 1/j, b - 1/j]$ is a countable union of closed intervals, so $\mathcal{B} \subset \mathcal{M}(\mathcal{E}_2)$ and hence $\mathcal{B}_{\mathbb{R}} = \mathcal{M}(\mathcal{B}) \subset \mathcal{M}(\mathcal{E}_2)$.

Case (c): Half-open intervals $\mathcal{E}_3 = \{(a, b] : a < b\}$

- $\mathcal{M}(\mathcal{E}_3) \subseteq \mathcal{B}_{\mathbb{R}}$: $(a, b] = (a, b) \cup \{b\}$ where $(a, b) \in \mathcal{B}_{\mathbb{R}}$ and $\{b\} = \cap_{j=1}^{\infty} (b - 1/j, b + 1/j) \in \mathcal{B}_{\mathbb{R}}$.
- $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_3)$: $(a, b) = \cup_{j=1}^{\infty} (a, b - 1/j]$ is a countable union of half-open intervals, so $\mathcal{B} \subset \mathcal{M}(\mathcal{E}_3)$ and hence $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_3)$.

Case (e): Closed rays $\mathcal{E}_7 = \{[a, \infty) : a \in \mathbb{R}\}$

- $\mathcal{M}(\mathcal{E}_7) \subseteq \mathcal{B}_{\mathbb{R}}$: $[a, \infty) = \cap_{j=1}^{\infty} (a - 1/j, \infty)$ is a countable intersection of open rays (generalized open intervals), so $\mathcal{E}_7 \subset \mathcal{B}_{\mathbb{R}}$.
- $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_7)$: $(a, b) = (a, \infty) \cap (-\infty, b)$ where $(a, \infty) = \cup_{j=1}^{\infty} [a + 1/j, \infty)$ and $(-\infty, b) = \mathbb{R} \setminus [b, \infty)$ are in $\mathcal{M}(\mathcal{E}_7)$, so $\mathcal{B} \subset \mathcal{M}(\mathcal{E}_7)$ and hence $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_7)$.

Cases (c'), (e') follow by symmetry.

5. We only need to show

$$\begin{aligned} \mathcal{B} &\subseteq \mathcal{M}(\mathcal{T}(\mathcal{B})) \\ \mathcal{T}(\mathcal{B}) &\subseteq \mathcal{M}(\mathcal{B}) \end{aligned} \tag{11}$$

- The former is trivial since $\mathcal{B} \subseteq \mathcal{T}(\mathcal{B}) \subseteq \mathcal{M}(\mathcal{T}(\mathcal{B}))$
- The later can be shown using (2). $\mathcal{M}(\mathcal{B})$ contains all countable unions of open intervals $\mathcal{B}$ while $\mathcal{T}(\mathcal{B})$ contains arbitrary union. (2) bridges the gap by claiming the arbitrary union can actually be replaced by countable union.

Construction of Measure

The big picture The general idea of constructing measure is, in crude terms:

1. Find a collection of set that has a clear notion of size
2. Approximate arbitrary subsets using element in the elementary set, hence get a notion of size on arbitrary subsets
3. Make the notion of size truly a measure by restricting to a special collection of subset.

Observe how the notion of size ripens:

$$\begin{aligned} \text{Stage 1: } \boxed{|\cdot| : \mathcal{E} \rightarrow \mathbb{R}^+} &\longrightarrow \text{Stage 2: } \boxed{\mu^* : P(X) \rightarrow \mathbb{R}^+} \\ &\longrightarrow \text{Stage 3: } \boxed{\mu : \mathcal{M}_{\mu^*} \rightarrow \mathbb{R}^+} \end{aligned} \tag{12}$$

1. *Stage 1 to 2*: Create a notion of size on arbitrary sets $A \subset P(X)$ by approximating them using “building blocks” ($A \in \mathcal{E}$) which has a notion of size $|\cdot|$. This sacrifices additivity and we instead of countable sub-additivity
2. *Stage 2 to 3*: Shrink the domain by restrict to a particular family $\mathcal{M}_{\mu^*} \subset P(X)$. This gives us *countable additivity* and μ is now a **complete measure**

Example 1.1

Concretely, think $\mathcal{E}$ as the set of all open rectangles in $\mathbb{R}^2$:

$$\mathcal{E} = \{(a, b) \times (c, d) | a < b, c < d; a, b, c, d \in \mathbb{R}\} \quad (13)$$

These clearly “canonical” notion of size, namingly:

$$|A| = (d - c)(b - a) \quad (14)$$

Now using these as “building blocks”, we can measure arbitrary sets $E \in P(\mathbb{R}^2)$, for instance, a unit circle. Here is a quick numerical demo for approximating a unit circle denoted by E with squares on a grid. The resolution n is defined as the number of intervals in $(0, 1)$.

As we take finer and finer squares, we approach a limit(or the liminf). This number can be used as a notion of size for the set E . This motivates the definition of *outer measure*.

How did we compute that?

A simple way is to count grid points that is inside the unit circle. A quick `julia` implementation gives the above plot:

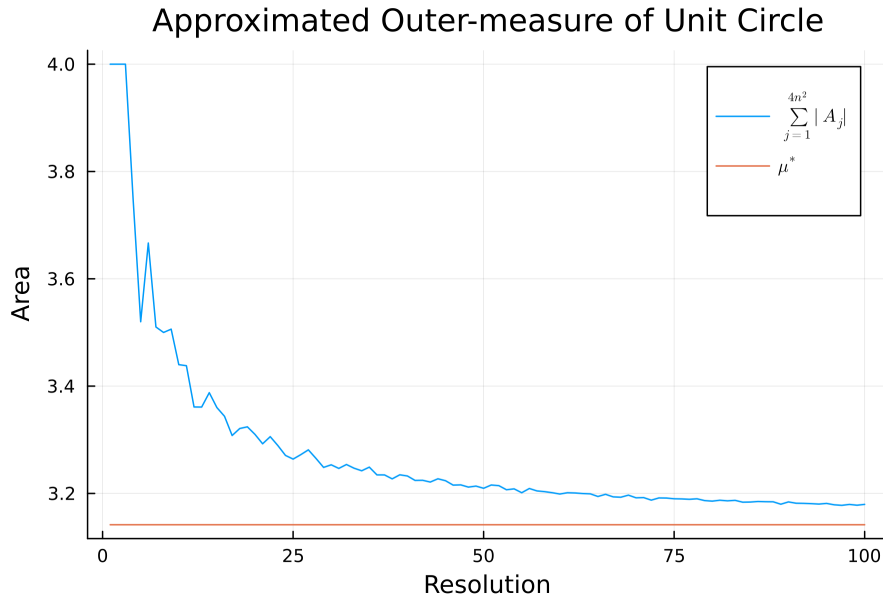
```
function compute_area_of_circle(n)
    cover::Int = 0
    for i in -n:n
        for j in -n:n
            if ((i/n)^2 + (j/n)^2) < 1
                if i == 0 && j == 0
                    cover += 4
                elseif i == 0 || j == 0
                    cover += 2
                else
                    cover += 1
                end
            end
        end
    end
    return cover * (1/n)^2
end
```

Outer Measure So how do we actually construct measures no matter what set we are working on?(Though in this book we are really interested in just $\mathbb{R}^n$). A general guideline is:

1. Find some *elementary set* $\mathcal{E}$ where premeasure is defined on the disjoint union of $\mathcal{E}$.
2. Define the measure of arbitrary subset $A \in P(X)$ as the the size of “smallest” cover of A using the elementary set:

$$\inf \left\{ \sum_{\nu=1}^{\infty} \rho(E_{\nu}) : E_{\nu} \in \mathcal{E}, A \subset \bigcup_{\nu=1}^{\infty} E_{\nu} \right\} \quad (15)$$

More generally we can define an *outer measure* as follows:



Definition 1.2 (Outer Measure)

An outer measure on a non-empty set X is a function $\mu^* : P(X) \rightarrow [0, \infty]$ such that:

- $\mu^*(\emptyset) = 0$
- If $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$
- $\mu^*(\cup_{\nu} A_{\nu}) \leq \sum_{\nu} \mu^*(A_{\nu})$

We note that property (2),(3) are both derived properties of measure. So we can think the definition of outer measure as a relaxation of measure: take out countable additivity and adds back two derived property. In addition, We note that (15) obeys Definition 1.1

Exercise

Show that (15) indeed defines an outer measure Definition 1.1.

Carathéodory Family $\mathcal{M}_{\mu^*}$ How do we then obtain measure from outer measure? We note that what outer measure lacks is finite additivity, so we can further ask, is there a collection of subset $\mathcal{M} \subset P(X)$ such that μ^* is actually finitely additive? The answer is actually YES! However this seems rather arbitrary:

$$\mathcal{M} = \{A \subset X : \mu^*(F) = \mu^*(F - A) + \mu^*(A \cap F), \forall F \subset X\} \quad (16)$$

We call elements of $\mathcal{M}$ **μ^* -measurable**. In plain English, this saying A can “split measure” arbitrary $F \subset X$ by measuring the intersection and the complement separately and add them together.

Theorem 1.1 (Carathéodory’s Theorem)

If μ^* is an outer measure of X then

1. the collection of **μ^* -measurable** sets:

$$\mathcal{M} = \{A \subset X : \mu^*(F) = \mu^*(A^c \cap F) + \mu^*(A \cap F), \forall F \subset X\} \quad (17)$$

is a σ -algebra.

2. $\mu^*|_{\mathcal{M}}$ is a **complete measure**

Premeasure and The Extension Theorem The workflow (12) is already very promising except one key issue:

Carathéodary Family can be Small

What if $\mathcal{M}_{\mu^*}$ is trivial or very small?

The next theorem prevents this by starting with an *algebra* $\mathcal{A}$ instead of just a plain family of sets $\mathcal{E}$. The plain notion of size on $\mathcal{E}$ is replaced by the so-called *premeasure*.

Definition 1.3 (Premeasure)

Suppose $\mathcal{A} \subset \mathcal{P}(X)$ is an algebra, a function $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ is a **premeasure** if

1. $\mu_0(\emptyset) = 0$
2. Let $\{A_j\}_1^\infty$ is a sequence of disjoint sets in $\mathcal{A}$. Given $\bigcup_1^\infty A_j \in \mathcal{A}$ then

$$\mu_0\left(\bigcup_1^\infty A_j\right) = \sum_1^\infty \mu_0(A_j) \quad (18)$$

Exercise 1.2 (Completeness of $\mathcal{M}_{\mu^*}$)

A measure μ on a σ -algebra $\mathcal{M}$ is complete if

$$\forall N \in \mathcal{M} \quad \forall N' \subseteq N \quad (\mu(N) = 0 \Rightarrow N' \in \mathcal{M}). \quad (19)$$

Let μ^* be an outer measure on a set X . Prove that the measure on $\mathcal{M}_{\mu^*}$ induced by μ^* by the Carathéodory construction is complete. (The proof of Carathéodory theorem in the book does not explain this well, write a detailed proof.)

Solution to Exercise 1.1

Proof 1.1

Note that $\mu(N) = 0 \implies \mu(N') = 0$. $\forall E \subseteq X$, by subadditivity:

$$\mu^*(E) \leq \mu^*(E \cap N') + \mu^*(E \cap N'^c) \quad (20)$$

By monotonicity and positivity of μ^* we have $\mu^*(E \cap N') = 0$ (since $E \cap N' \subseteq N'$ so $\mu^*(E \cap N') \leq 0$). But again by monotonicity $\mu^*(E \cap N'^c) \leq \mu^*(E)$ hence we have:

$$\mu^*(E) \leq \mu^*(E \cap N') + \mu^*(E \cap N'^c) \leq \mu^*(E) \quad (21)$$

Hence:

$$\mu^*(E) = \mu^*(E \cap N') + \mu^*(E \cap N'^c) \quad (22)$$

As desired.

Exercise 1.3

Let $\mathcal{A} \subset \mathcal{P}(X)$ be an algebra, $\mathcal{A}_\sigma$ the collection of countable unions of sets in $\mathcal{A}$, and $\mathcal{A}_{\sigma\delta}$ the collection of countable intersections of sets in $\mathcal{A}_\sigma$. Let μ_0 be a premeasure on $\mathcal{A}$ and μ^* the induced outer measure.

1. For any $E \subset X$ and $\epsilon > 0$ there exists $A \in \mathcal{A}_\sigma$ with $E \subset A$ and $\mu^*(A) \leq \mu^*(E) + \epsilon$.

2. If $\mu^*(E) < \infty$, then E is μ^* -measurable iff there exists $B \in \mathcal{A}_{\sigma\delta}$ with $E \subset B$ and $\mu^*(B \setminus E) = 0$.
3. If μ_0 is σ -finite, the restriction $\mu^*(E) < \infty$ in (2) is superfluous.

Solution to Exercise 1.1

1. This is just the definition of outer measure (namely, infimum). Note since

$$\mu^*(E) = \inf C \quad (23)$$

where

$$C = \left\{ \sum \mu_0(A_j) \mid A = \cup A_j \in \mathcal{A}_\sigma, E \subset A \right\} \quad (24)$$

Hence $\exists c \in C$ such that $c < \mu^*(E) + \epsilon$ by the definition of infimum. Now the map between open covers and their area is clearly surjective by construction:

$$\begin{aligned} \{A \in \mathcal{A}_\sigma : E \subset A\} &\rightarrow \mathbb{R} \\ A &\mapsto \underbrace{\sum_j \mu_0(A_j)}_{\mu^*(A)} \end{aligned} \quad (25)$$

hence we can find $A \in \mathcal{A}_\sigma$ that corresponds to c .

3. ($\implies$) Suppose E is μ^* -measurable. Now by (1), for each n we can find $A_n \in \mathcal{A}_\sigma$ with $E \subset A_n$ and $\mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$. The A_n need not be decreasing, so replace them by $\tilde{A}_n := \bigcap_{k=1}^n A_k$; since $\mathcal{A}_\sigma$ is closed under finite intersections (intersection distributes over the countable unions defining each A_k), $\tilde{A}_n \in \mathcal{A}_\sigma$, and now $\tilde{A}_n$ is a decreasing sequence with $E \subset \tilde{A}_n \subset A_n$, so still $\mu^*(E) \leq \mu^*(\tilde{A}_n) \leq \mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$. Define:

$$B := \bigcap_n \tilde{A}_n = \bigcap_n A_n \in \mathcal{A}_{\sigma\delta} \quad (26)$$

By construction $E \subset B$. Since $\tilde{A}_n \in \mathcal{A}_\sigma \subset \mathcal{M}_{\mu^*}$ is a decreasing sequence with $\mu^*(\tilde{A}_1) \leq \mu^*(E) + 1 < \infty$, *continuity-from-above* on $\mathcal{M}_{\mu^*}$ gives:

$$\mu^*(B) = \lim_{n \rightarrow \infty} \mu^*(\tilde{A}_n) = \mu^*(E) \quad (27)$$

Now use that E is μ^* -measurable, testing it against B : since $E \subset B$,

$$\mu^*(B) = \mu^*(B \cap E) + \mu^*(B \cap E^c) = \mu^*(E) + \mu^*(B \setminus E) \quad (28)$$

Combining with $\mu^*(B) = \mu^*(E) < \infty$ and subtracting the finite common value $\mu^*(E)$ from both sides gives $\mu^*(B \setminus E) = 0$, as desired.

($\impliedby$) Suppose there exists $B \in \mathcal{A}_{\sigma\delta}$ such that $E \subset B$ and $\mu^*(B \setminus E) = 0$. Note any $B \in \mathcal{A}_{\sigma\delta}$ is μ^* -measurable since $\mathcal{A} \subset \mathcal{M}_{\mu^*}$ (by *extension theorem*). We would like for all $F \subset X$

$$\mu^*(F) \geq \mu^*(F \cap E) + \mu^*(F \cap E^c) \quad (29)$$

But now $E \subset B$ hence:

$$\begin{aligned} \mu^*(F) &= \mu^*(F \cap B) + \mu^*(F \cap B^c) \\ &\geq \mu^*(F \cap E) + \mu^*(F \cap B^c) \end{aligned} \quad (30)$$

Note all we need now is $\mu^*(F \cap B^c) \geq \mu^*(F \cap E^c)$.

But now

$$\mu^*(F \cap E^c) = \mu^*(F \cap (B^c \cup (B \cap E^c))) \leq \mu^*(F \cap B^c) + \mu^*(F \cap (B \cap E^c)) \quad (31)$$

Again, by monotonicity and the assumption

$$\mu^*(F \cap (B \cap E^c)) \leq \mu^*(B \cap E^c) = 0 \quad (32)$$

Therefore $\mu^*(F \cap B^c) \geq \mu^*(F \cap E^c)$. As desired.

4. Suppose μ_0 is σ -finite then $\exists \{X_n\}$, $\mu_0(X_n) < \infty$ such that $X = \cup_n X_n$. Now define $E_n = X_n \cap E$, we can apply (2) to each E_n (There existss $B_n \in \mathcal{A}_{\sigma\delta}$ satisfying the property iff $E_n \in \mathcal{M}$).

($\implies$) Say we already have desired set B_n for each E_n (each $B_n \in \mathcal{A}_{\sigma\delta}$, $E_n \subset B_n$, $\mu^*(B_n \setminus E_n) = 0$)^a. Take the union $B = \cup_n B_n$; it covers E since $E_n \subset B_n$ and $E = \cup_n E_n$. It also satisfies the null-excess property: since $E_n \subset E$ we have $B_n \setminus E \subset B_n \setminus E_n$ for each n , so $B \setminus E = \cup_n (B_n \setminus E) \subset \cup_n (B_n \setminus E_n)$, and by countable subadditivity:

$$\mu^*(B \setminus E) \leq \sum_n \mu^*(B_n \setminus E_n) = \sum_n 0 = 0 \quad (33)$$

($\impliedby$) Suppose there is such B for E (possibly of infinite outer-measure). Define $E_n = X_n \cap E$ as before, and now also $B_n := X_n \cap B \in \mathcal{A}_{\sigma\delta}$ (intersecting with $X_n \in \mathcal{A}$ preserves membership in $\mathcal{A}_{\sigma\delta}$). Then $E_n \subset B_n$, and

$$\mu^*(B_n \setminus E_n) = \mu^*(X_n \cap (B \setminus E)) \leq \mu^*(B \setminus E) = 0 \quad (34)$$

Since $\mu^*(E_n) \leq \mu_0(X_n) < \infty$, part (2) applies to E_n and gives $E_n \in \mathcal{M}$. Now we want to show $E \in \mathcal{M}$ but this is trivial as $\mathcal{M}$ is a σ -algebra closed under countable union and $E = \cup_n E_n$

^aStrictly, $\mathcal{A}_{\sigma\delta}$ need not be closed under *countable* unions (only finite ones) — the analogous fact for G_δ sets in $\mathbb{R}$ fails: $\mathbb{Q} = \bigcup_q \{q\}$ is a countable union of G_δ singletons yet is not itself G_δ . So $B = \cup_n B_n$ below should be read as living in $\mathcal{M}_{\mu^*}$ (a genuine σ -algebra), which is all that is actually used afterward.

Exercise 1.4

Let μ^* be an outer measure on X induced from a finite premeasure μ_0 . If $E \subset X$, define the inner measure of E to be $\mu_*(E) = \mu_0(X) - \mu^*(E^c)$. Then E is μ^* -measurable iff $\mu^*(E) = \mu_*(E)$

Solution to Exercise 1.1

($\implies$) Suppose E is μ^* -measurable. Then so is E^c , and since $X = E \sqcup E^c \in \mathcal{A}$, additivity gives $\mu_0(X) = \mu^*(X) = \mu^*(E) + \mu^*(E^c)$, i.e. $\mu^*(E) = \mu_*(E)$.

($\impliedby$) Suppose $\mu^*(E) + \mu^*(E^c) = \mu_0(X) = \mu^*(X) < \infty$. Fix $\epsilon > 0$; by Exercise 1.1 part (1), pick $A, A' \in \mathcal{A}_\sigma \subset \mathcal{M}_{\mu^*}$ with $E \subset A$, $E^c \subset A'$, $\mu^*(A) \leq \mu^*(E) + \epsilon$, $\mu^*(A') \leq \mu^*(E^c) + \epsilon$. Then $A \cup A' = X$, so

$$\mu^*(X) + \mu^*(A \cap A') = \mu^*(A) + \mu^*(A') \leq \mu^*(X) + 2\epsilon \implies \mu^*(A \cap A') \leq 2\epsilon \quad (35)$$

Since $A \setminus E \subset A \cap A'$, $\mu^*(A \setminus E) \leq 2\epsilon$. Taking $\epsilon = \frac{1}{n}$ gives $A_n \supset E$ with $\mu^*(A_n \setminus E) \leq \frac{2}{n}$; let $B := \bigcap_n A_n \in \mathcal{M}_{\mu^*}$. Then $E \subset B$ and $\mu^*(B \setminus E) \leq \mu^*(A_n \setminus E) \rightarrow 0$, so $\mu^*(B \setminus E) = 0$.

This is exactly the hypothesis of the ($\impliedby$) direction of Exercise 1.1 part (2) (only measurability of B was used there, not $B \in \mathcal{A}_{\sigma\delta}$), which concludes E is μ^* -measurable. As desired.

The following exercise is assumed everywhere in mathematics and is not directly related to outer measure. But they are invoked multiple times, for instance, in subadditivity of outer measure.

Exercise 1.5 (Order invariance for Series of Positive Terms)

1. Prove that any generalized series with summands in $[0, \infty]$ can be summed in any order. Specifically, if $a_j \in [0, \infty]$ for $j \in \mathbb{N} = \{1, 2, 3, \dots\}$, then for any bijection $\mathbb{N} \rightarrow \mathbb{N}$, $j \mapsto k_j$, the equality $\sum_{j=1}^{\infty} a_j = \sum_{j=1}^{\infty} a_{k_j}$ holds. (That is, a nonnegative generalized series can be summed in any order.)
2. Prove that any two generalized series with summands in $[0, \infty]$ can be summed as one series. Specifically, if $a_j, b_j \in [0, \infty]$ for $j \in \mathbb{N}$, then the equality $\sum_{j=1}^{\infty} a_j + \sum_{j=1}^{\infty} b_j = \sum_{j=1}^{\infty} (a_j + b_j)$ holds.

Solution to Exercise 1.1

1. We first make a claim

$$\sum_{j \in \mathbb{N}} a_j = \sup_J \sum_{j \in J} a_j \quad (36)$$

where $J \subset \mathbb{N}, |J| < \infty$.

- To show $\sum_{j \in \mathbb{N}} a_j \leq \sup_J \sum_{j \in J} a_j$:

$$\sum_{j \in \mathbb{N}} a_j = \lim_{N \rightarrow \infty} \sum_{j=1}^N a_j \leq \sup_J \sum_{j \in J} a_j \quad (37)$$

- To show $\sum_{j \in \mathbb{N}} a_j \geq \sup_J \sum_{j \in J} a_j$, we first note $\forall J$ as above:

$$\sum_{j \in \mathbb{N}} a_j = \sup_m s_m \geq s_{\max J} \geq \sum_{j \in J} a_j \quad (38)$$

Now sup preserves non-strict inequality so we can take sup on both sides to get:

$$\sup_J \sum_{j \in \mathbb{N}} a_j = \sum_{j \in \mathbb{N}} a_j \geq \sup_J \sum_{j \in J} a_j \quad (39)$$

as desired. Now (1) becomes trivial since $\forall J$ defined as above, suppose we define:

$$k(J) = \{k(j) : j \in J\} \quad (40)$$

now by the claim

$$\sum_{j \in \mathbb{N}} a_{k(j)} = \sup_J \sum_{j \in J'} a_j = \sup_J \sum_{j \in k(J)} a_j \quad (41)$$

where the last equality follows as J' is also in the set $\{J \subset \mathbb{N} : |J| < \infty\}$.

1. We first make a claim: For monotone non-decreasing sequences s_n, t_n :

$$\sup(s_n + t_n) = \sup s_n + \sup t_n \quad (42)$$

- $\sup(s_n + t_n) \leq \sup s_n + \sup t_n$ is true since $\forall m, s_m + t_m \leq \sup s_n + \sup t_n$ and taking $\sup_m$ on both sides get us desired.

- $\sup(s_n + t_n) \geq \sup s_n + \sup t_n$ can be shown by arguing $\forall \epsilon > 0$,

$$\sup(s_n + t_n) \geq \sup s_n + \sup t_n - \epsilon \quad (43)$$

Denote $s = \sup s_n, t = \sup t_n$. By definition of supremum, we can pick N_1, N_2 such that $s_{N_1} > s - \epsilon/2$ and $t_{N_2} > t - \epsilon/2$. Let $N = \max(N_1, N_2)$; since s_n, t_n are non-decreasing, $\forall n \geq N$ we have $s_n \geq s_{N_1} > s - \epsilon/2$ and $t_n \geq t_{N_2} > t - \epsilon/2$, hence:

$$s_n + t_n > s + t - \epsilon \quad (44)$$

Taking $\sup_n$ on the left gives $\sup(s_n + t_n) \geq s + t - \epsilon = \sup s_n + \sup t_n - \epsilon$. As desired, since $\epsilon > 0$ was arbitrary. (2) then easily follows by taking the claim in (1) and then let

$$\begin{aligned} s_n &= \sum_{j=1}^n a_j \\ t_n &= \sum_{j=1}^m b_j \end{aligned} \quad (45)$$

Apply the above claim gets desired.

Casestudy: Lebesgue-Stieltjes Measure

The most useful set that we would like to equip a measure with is, unarguably, the real numbers $\mathbb{R}$.

Starting Point: The half-intervals Recall that we would start with an elementary collection that has a clear notion of size. We define the **h-intervals**:

$$H = \{(a, b] : a < b, a, b \in \mathbb{R}\} \quad (46)$$

We define a terminology here:

Definition 1.4 (Borel Sets)

Let X any metric space, the σ -algebra generated by the family of open sets in X is called the **Borel σ -algebra**, denoted by $\mathcal{B}_X$

Exercise 1.6 (h-intervals generate $\mathcal{B}_{\mathbb{R}}$)

Check H is an elementary set and so it generates an algebra $\mathcal{A}$. Furthermore, show that the σ -algebra generated by $\mathcal{A}$ is $\mathcal{B}_{\mathbb{R}}$

Another terminology:

Definition 1.5 (Elementary Decomposition)

Suppose $A \in \mathcal{A}$, since $\mathcal{A} = \mathcal{M}(H)$ we can find finite disjoint h-intervals $\{I_j\}_{j=1}^N = \{(a_j, b_j]\}_{j=1}^N$ such that:

$$A = \bigcup_{j=1}^N I_j \quad (47)$$

Then $\{I_j\}$ is an **elementary decomposition** of A .

We define size on H as follows:

Notion of Size on h-intervals

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and right-continuous, let $(a, b] \in H$, then:

$$|(a, b]| = F(b) - F(a) \quad (48)$$

Note when $F(x) = x$, we get the common notion of length of interval.

The next step is to extend this notion of size to $\mathcal{A} = \mathcal{M}(H)$.

Step 1: Construction of a premeasure Now we have an algebra $\mathcal{A}$ where elements of the algebra $A \in \mathcal{A}$ has an elementary decomposition $A = \{I_j\}$ to h-intervals $I = (a_j, b_j]$, which is equipped with a notion of size $b_j - a_j$. A natural way to define the notion of size for A is hence:

$$\mu_0(A) = \sum_{j=1}^N |I_j| = \sum_{j=1}^N F(b_j) - F(a_j) \quad (49)$$

We claim that μ_0 is actually a *pre-measure* on $\mathcal{A}$.

Proposition 1.1

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ to be a right-continuous, increasing function. Define μ_0 as

$$\begin{aligned} \mu_0 : \mathcal{A} &\rightarrow \mathbb{R} \\ \bigcup_{j=1}^N A &\mapsto \sum_{j=1}^N [F(b_j) - F(a_j)] \end{aligned} \quad (50)$$

where $\{I_j\} = \{(a_j, b_j]\}$ is an elementary decomposition. Then μ_0 is a premeasure on $\mathcal{A}$.

Proof 1.2 (μ_0 is a pre-measure)

We first need to show that μ_0 is actually well-defined^a. Then we illustrate:

1. Empty set has zero measure
2. Non-negativity
3. Finite additivity
4. Countable additivity where only (1), (2), (4) is truly required. But (3) is used in the proof of (4).

^aNote that in defining a notion of size μ_0 on $A \in \mathcal{A}$, we requires A to have a concrete representation $\cup_j I_j$. If the result is independent on how we choose this representation(i.e. different elementary decomposition $\{I_j\}$), then this notion of size is indeed fundamentally correct. Otherwise if μ_0 is dependent on how we choose representation, then it is not a credible notion of size. Such “representational invariance” is called *well-definedness* in mathematics.

1.2 Measurable Functions

CHAPTER II

Vector Space

2.1 Finite-dimensional Vector Space

Linear Operators

Change of basis Let V be a finite-dimensional vector space of dimension n . Let $\beta = \{e_1, \dots, e_n\}$ and $\beta' = \{e'_1, \dots, e'_n\}$ be both basis for V . We are interested in change representation of $x \in V$ from using β to β' . To do this we need to know how $e_j \in \beta$ can be represented using β' :

$$e_\nu = a_\nu^\mu e'_\mu \quad (51)$$

Here coefficients a_ν^μ can be organized into a matrix. The matrix a_ν^μ looks like:

$$\begin{pmatrix} a_1^1 & a_2^1 & \cdots & a_n^1 \\ \vdots & \vdots & & \vdots \\ a_1^n & a_2^n & \cdots & a_n^n \end{pmatrix} \quad (52)$$

Contravariant Transformation There are two interpretations of a_ν^μ :

1. (a_ν^μ transform β' to β) This is the definition above:

$$e_\nu = a_\nu^\mu e'_\mu \quad (53)$$

2. (a_ν^μ transform representation in β to β') Now for arbitrary $x \in V$, we have:

$$x = c'_\nu e'_\nu = c_\nu e_\nu = c_\nu (a_\nu^\mu e'_\mu) = (c_\nu a_\nu^\mu) e'_\mu \quad (54)$$

from which we recognize:

$$c'_\nu = (c_\nu a_\nu^\mu) \quad (55)$$

We note that the matrix has *opposite* effect on basis and representation. This is actually intuitive: in order to represent the vector in new basis β' , we will have to write the old basis $e_\nu \in \beta$ using linear combinations of the new basis e'_μ . The later which effectively transforms basis in β' to β . For this reason, the representation is said to be transformed **contravariantly**.

Notation

The matrix a_ν^μ is commonly denoted by $[I]_\beta^{\beta'}$ in math textbooks. Therefore:

- Transform in representation is denoted by:

$$[x]_{\beta'} = [I]_\beta^{\beta'} [x]_\beta \quad (56)$$

- Transform in matrix representation of operator is given by:

$$[T]_\beta = [I]_\beta^{\beta'} [T]_{\beta'} [I]_\beta^{\beta'} \quad (57)$$

Triangularization

Proposition 2.1 (Existence of eigenvalue in vector spaces on algebraically closed field)

Suppose V is a finite-dimensional vector space over an algebraically closed field $\mathbb{F}$, and $T \in \mathcal{L}(V)$, then T has at least one eigenvalue

Theorem 2.1 (Algebraically closed field guarantees triangularizability)

Suppose V is a finite-dimensional vector space over an algebraically closed field $\mathbb{F}$, and $T \in \mathcal{L}(V)$, then T is always triangularizable.

Inner Product Space

Note: The development of concepts need to keep extension to infinite analog in mind

The identity operator and the Kroncker's Delta Let V be a vector space endorsed with an inner product $\langle \cdot | \cdot \rangle$ of dimension n and $\beta = \{e_\nu\}$ be an orthonormal basis. Then we can compute the inner product between arbitrary $x, y \in V$ as follows:

$$\begin{aligned}\langle x|y\rangle &= \langle e_\nu|x\rangle^* \langle e_\nu|y\rangle \\ &= \langle x|e_\nu\rangle \langle e_\nu|y\rangle \\ &= \langle x||e_\nu\rangle \langle e_\nu||y\rangle\end{aligned}\tag{58}$$

But $\langle x|y\rangle = \langle x|I|y\rangle$, hence we get that

Lemma 2.1 (Outer-product decomposition of the identity operator)

Let $\{e_\nu\}$ be an orthonormal basis for V , then:

$$|e_\nu\rangle\langle e_\nu| = I\tag{59}$$

Help! What is happening here?

Step 1 We first rewrite inner product as a sum of component-component product. In $\mathbb{C}^3$ for instance,

$$\begin{aligned}\langle x|y\rangle &= \sum_\nu (x^\nu)^* y^\nu \\ &= \sum_\nu \langle e_\nu|x\rangle^* \langle e_\nu|y\rangle\end{aligned}\tag{60}$$

Note in the last step we envicted orthonormality of $\{e_\nu\}$. Adopting Einstein notaiton we get the first equality.

Step 2 We are bascially writing two consecutive inner product $\langle x, \alpha \rangle \langle \beta, y \rangle$ as a bilinear form $\langle x, Ay \rangle$ where $A = |\alpha\rangle\langle\beta|$. Consider this example in $\mathbb{R}^{\neq}$:

$$\begin{aligned}\langle x, \alpha \rangle \langle \beta, y \rangle &= \begin{pmatrix} x^1 & x^2 & x^3 \end{pmatrix} \begin{pmatrix} \alpha^1 \\ \alpha^2 \\ \alpha^3 \end{pmatrix} \begin{pmatrix} \beta^1 & \beta^2 & \beta^3 \end{pmatrix} \begin{pmatrix} y^1 \\ y^2 \\ y^3 \end{pmatrix} \\ &= \begin{pmatrix} x^1 & x^2 & x^3 \end{pmatrix} \begin{pmatrix} \alpha^1\beta^1 & \dots & \alpha^1\beta^3 \\ \vdots & & \vdots \\ \alpha^3\beta^1 & \dots & \alpha^3\beta^3 \end{pmatrix} \begin{pmatrix} y^1 \\ y^2 \\ y^3 \end{pmatrix}\end{aligned}\tag{61}$$

In otherword, we are just invoking associative law(which holds in vector space):

$$(x\alpha^T)(\beta y^T) = x(\alpha^T\beta)y^T\tag{62}$$

where the outer product $\alpha^T\beta = A$.

We denote bilinear forms $x^T Ay$ in sandwich notations: $\langle x|A|y\rangle$

(Step 3) The only possible opeartor to be sandwiched without changing the result:

$$\langle x|Ay\rangle = \langle x|y\rangle\tag{63}$$

is when $A = I$. We therefore conclude that the outer product $|e_\nu\rangle\langle e_\nu|$ is the identity. In math notation, this is saying:

- For orthonormal basis $\{e_\nu\}$, we have

$$\sum_{\nu=1}^n e_\nu e_\nu^T = I\tag{64}$$

Now we consider representing $|x\rangle$ in β :

$$\begin{aligned} |x\rangle &= (|e_\nu\rangle\langle e_\nu|)|x\rangle \\ &= |e_\nu\rangle\langle e_\nu|x\rangle \end{aligned} \tag{65}$$

Now consider taking the inner product with another basis e_μ :

$$\langle e_\mu|x\rangle = \langle e_\mu|e_\nu\rangle\langle e_\nu|x\rangle \tag{66}$$

2.2 Infinite-dimensional Vector Space

Linear Operators

L^p Space

Definition 2.1 (L^p space)

Let (X, Σ, μ) be a measure space and $1 \leq p \leq \infty$. The space $L^p(X, \mu)$ is the set of all measurable functions from X to $\mathbb{R}$ or $\mathbb{C}$ such that $\int_X |f(x)|^p d\mu < \infty$.

- Note L^p is a *normed space* with norm:

$$\|f\|_{L^p} = \left(\int_X |f|^p d\mu \right)^{1/p} \quad (67)$$

p can be the symbol ∞ with the norm defined as:

$$\|f\|_{L^\infty} = \inf\{C \in \mathbb{R}_{\geq 0} : |f(x)| < C, \forall x\} \quad (68)$$

- It is *complete*, hence it is a Banach Space

Linear Operators

Definition 2.2 (Linear Operators)

Let T be a map between two normed space X and Y . T is linear if:

$$T(x + \lambda y) = T(x) + \lambda T(y) \quad (69)$$

$$\forall x, y \in X, \lambda \in K$$

If we consider linear functions $f : \mathbb{R} \rightarrow \mathbb{R}$, it seems continuity is free since we can always write $T(x) = kx$ hence:

$$\|T(x) - T(x_0)\| = \|k(x - x_0)\| = k\|x - x_0\| \quad (70)$$

so we can easily find a delta ball of radius $\frac{\epsilon}{k}$ given an epsilon ball such that the delta ball is mapped into the epsilon ball. More generally we can define bounded operator to make the above logic work:

Definition 2.3 (Bounded Operators)

Let T be a map between two normed space X and Y . T is bounded if $\exists C > 0$:

$$\|T(x)\|_Y \leq C\|x\|_X \quad (71)$$

$$\forall x \in X$$

Proposition 2.2 (Continuous and Bounded are equivalent for Linear Operator)

The following are equivalent:

1. If X and Y are normed spaces and $T : X \rightarrow Y$ is a linear map which is continuous from X to Y
2. T is bounded

Proof 2.1

- Suppose T is **linear and bounded**, $\forall x_0 \in X$, $\forall \epsilon > 0$, consider the open ball around x_0 with radius $\epsilon \in B_\epsilon^o$. Then $\forall x \in B_\epsilon^o$:

$$\begin{aligned} \|T(x) - T(x_0)\| &= \|T(x - x_0)\| \\ &\leq C\|x - x_0\| \end{aligned} \quad (72)$$

Now to make $C\|x - x_0\| < \epsilon$, we need $\|x - x_0\| < \frac{\epsilon}{C}$. Hence $\delta(\epsilon) = \frac{\epsilon}{C}$

- Suppose T is **linear and continous**. Then in particular T is continous at $x = 0$. We can find a $\delta > 0$ such that $\|x\| < \delta \Rightarrow \|T(x)\| < 1$. Now to control arbitrary $x \in X$, we can rewrite x as

$$x = \left(\frac{2}{\delta}\right) \|x\| \left(\frac{\delta x}{2\|x\|}\right) = C\|x\|x' \quad (73)$$

Note $\|x'\| = \frac{\delta}{2}\|x\| < \delta$ hence $\|T(x')\| \leq 1$. By linearity $T(x) = T(C\|x\|x') = C\|x\|T(x')$. Thus:

$$\|T(x)\|_Y = C\|x\|\|T(x')\| \leq C\|x\| \quad (74)$$

It turns out all finite-dimensional linear operator are bounded. However, infinite-dimensional linear operator can easily be unbounded.

Example 2.1 (Unbounded Opeartor)

Consider the **differential operator** $\frac{d}{dx} : C^\infty[0, 2\pi] \rightarrow C^\infty[0, 2\pi]$ Consider the family $f_n = \sin(nx)$. Now

$$\begin{aligned} \|f_n\|_{L_2}^2 &= \int_0^{2\pi} \sin^2(nx) dx \\ &= \int_0^{2\pi} \frac{1 - \cos(2x)}{2} dx \\ &= \pi \end{aligned} \quad (75)$$

However $\frac{df_n}{dx} = n \cos(nx)$ and:

$$\begin{aligned} \|f'_n\|_{L_2}^2 &= \int_0^{2\pi} n^2 \cos^2(nx) dx \\ &= n^2 \int_0^{2\pi} \frac{1 + \cos(2x)}{2} dx \\ &= n^2 \pi \end{aligned} \quad (76)$$

Hence the operator $\frac{d}{dx}$ is not bounded since suppose there is such C :

$$\|f'\|^2 \leq C\|f\|^2 \quad (77)$$

We can always pick n , $n^2 > C$ such that $n^2\pi = \|f'_n\|^2 > C\|f_n\|^2 = C\pi$.

The Uniform Boundedness Theorem

Baire Category Theorem

Theorem 2.2 (Baire Category Theorem)

Let X be a **complete metric space** then:

1. Open dense subsets of X is closed under *countable union*
2. X is not a countable union of nowhere dense set

Linear Differential Operator

The Formal Linear Differential Operator We define the differential operator as:

Definition 2.4 (Differentiation Operator)

Let $D : L^2[a, b] \rightarrow L^2[a, b]$ be the mapping defined by:

$$D(u) = \frac{d}{dx}u \quad (78)$$

for all $u \in L^2[a, b]$

Now we can define the *formal linear operator* by some operator algebra:

Definition 2.5 (Formal linear differential operator)

Let $p(x)$ be any polynomial over $\mathbb{C}$ of positive degree:

$$p(x) = a_\nu x^\nu \quad (79)$$

where a_ν are functions of x . Define $L := p(D)$ as the **formal linear differential operator** that correspond to the **auxillary polynomial** p :

$$L = a_\nu D^\nu \quad (80)$$

The **order** of L is given by the order of p .

We call it *formal* since we want to contrast with *concrete* linear differential operator which is a formal operator endorsed with boundary conditions (This convention is used in *Mathematical Physics* by Michael Stone and Paul Goldbart). We can think of the difference between them as:

- The **nullspace of formal operator** is the space of general solution of homogenous differential equation
- The **nullspace of concrete operator** is the space of particular solution to homogenous differential equation with boundary conditions

We'll explore the nullspace of formal operators with a bit more detail now.

Nullspace of formal operator We first observe something very nice about solutions to homogenous differential equations: they belong to C^∞ (infinitely differentiable).

The Bootstrapping Lemma

Lemma 2.2

Let L be a formal operator of order n . Suppose the coefficients of the auxillary polynomial are k times continuously differentiable ($a_\nu \in C^k$) then $\forall y \in \text{Null}(L)$, y is at least $k + n$ times continuously differentiable.

Proof 2.2

Suppose $y \in \text{Null}(L)$, note trivially $y \in C^n$. Then:

$$\begin{aligned} a_0 y + a_1 y^{(1)} + a_2 y^{(2)} \dots + a_n y^{(n)} &= 0 \\ y^{(n)} &= -(b_0 y + b_1 y^{(1)} + b_2 y^{(2)} \dots \underbrace{b_{n-1} y^{(n-1)}}_{\in C^{\min(k,1)}}) \end{aligned} \quad (81)$$

Where $b_\nu = a_\nu / a_n$. Now $b_\nu \in C^k$ by closedness under polynomial division. Also $y^{(\nu)} \in C^{n-\nu}$ hence the R.H.S is at least one time continuously differentiable (the bottle neck being $a_{n-1} y^{(n-1)} \in$

$C^{\min(k,1)}$). Therefore $y^{(n)} \in C^1$. But this implies $y \in C^{n+1}$ hence we repeat the above argument to get $y^{(\nu)} \in C^{n+1-\nu}$. The R.H.S has the bottleneck $a_{n-1}y^{(n-1)} \in C^{\min(k, n+1-(n-1))} = C^{\min(k,2)}$. We can repeat this procedure m times and yield that $y^{(n)} \in C^{\min(k,m)}$. This means the best we can get is $y^{(n)} \in C^k$ or $y \in C^{n+k}$ (after which the coefficient becomes the bottleneck).

Corollary 2.1

If the coefficients are in C^∞ then $y \in \text{Null}(L)$ are in C^∞ .

Nullspace of L is an n -dimensional subspace of C^∞

Linear Independence: Wronskian

Solving $Ly = 0$ with constant coefficients

$Ly = 0$ with general $a_\nu(x)$

Exercise 2.1 (1-dimensional Scattering Problem)

Consider the 1d Schrodinger equation:

$$-\frac{d^2\psi}{dx^2} + V\psi = E\psi \quad (82)$$

This is an eigenvalue problem of the operator $H = -\frac{d^2}{dx^2} + V$:

$$H\psi = E\psi \quad (83)$$

Where V has finite support $[-a, a]$, $a \in \mathbb{R}$. Denote:

1. $L := \{x \in \mathbb{R} : x < -a\}$
2. $R := \{x \in \mathbb{R} : x > a\}$

Problems:

1. Show that in L, R , the general solution to the formal operator $H - E$ where $E = k^2$ is:

$$\begin{aligned} \psi_k &= A \exp(ikx) + B \exp(-ikx) & x \in L \\ \psi_k &= C \exp(ikx) + D \exp(-ikx) & x \in R \end{aligned} \quad (84)$$

2. Now consider two asymptotic boundary condition:

$$\begin{cases} \Pi_L : D = 0, x \rightarrow \infty \\ \Pi_R : C = 0, x \rightarrow -\infty \end{cases} \quad (85)$$

where Π_L depicts a left-incident wave with no incoming wave from the right and Π_R is the opposite. Show that the solution(in L, R) for the BC Π_1 and $k > 0$ is:

$$\psi_k^L(x) = \begin{cases} \exp(ikx) + r_L(k) \exp(-ikx) & x \in L \\ t_L(k) \exp(ikx) & x \in R \end{cases} \quad (86)$$

Similarly, for Π_2 , $k < 0$:

$$\psi_k^R(x) = \begin{cases} t_R(k) \exp(ikx) & x \in L \\ \exp(ikx) + r_R(k) \exp(-ikx) & x \in R \end{cases} \quad (87)$$

where t, r denotes the *transmission* and *reflection* coefficient.

3. Note that ψ_k^* is also a solution. Use properties of Wronskian to show that:

$$|t_{L/R}|^2 + |r_{L/R}|^2 = 1 \quad (88)$$

Solution to Exercise 2.2

1. Fix a k . Consider the basis ψ_k^L, ψ_k^{L*} . Form the Wronskian in L :

$$W_L = \begin{vmatrix} \exp(ikx) + r_L \exp(-ikx) & \exp(-ikx) + r_L^* \exp(ikx) \\ ik \exp(ikx) - ikr_L \exp(-ikx) & -ik \exp(-ikx) + ikr_L^* \exp(ikx) \end{vmatrix} \quad (89)$$

Evaluate the determinant:

$$\begin{aligned} W_L &= (-ik) - ikr_L \exp(-2ikx) + ikr_L \exp(2ikx) + ik|r_L|^2 \\ &\quad - [ik - ikr_L \exp(-2ikx) + ikr_L \exp(2ikx) - ik|r_L|^2] \\ &= 2ik(|r_L|^2 - 1) \end{aligned} \quad (90)$$

Next form Wronskian in R :

$$W_R = \begin{vmatrix} t_L \exp(ikx) & t_L \exp(-ikx) \\ ikt_L \exp(ikx) & -ikt_L \exp(-ikx) \end{vmatrix} = -2ik|t_L|^2 \quad (91)$$

Since we would expect $W_R = W_L = 0$, hence:

$$\begin{aligned} W_R - W_L &= 0 \\ 2ik(|r_L|^2 + |t_L|^2 - 1) &= 0 \end{aligned} \quad (92)$$

Since $k > 0$ we get as desired.

4. Fix k . We repeat the same procedure with $\{\psi_k^L, \psi_{(-k)}^R\}$. We evaluate W_L :

$$\begin{aligned} W_L &= \begin{vmatrix} \exp(ikx) + r_L(k) \exp(-ikx) & t_R(-k) \exp(-ikx) \\ ik \exp(ikx) - ikr_L(k) \exp(-ikx) & -ikt_R(-k) \exp(-ikx) \end{vmatrix} \\ &= -2ikt_R(-k) \end{aligned} \quad (93)$$

Similarly,

$$W_R = -2ikt_L(k) \quad (94)$$

Again, $W_L = W_R = 0$ so we get as desired.

Hilbert Space

In short, Hilbert Space is a complete inner-product space that is infinite-dimensional.

Definition 2.6 (Hilbert Space)

Let X be a complex vector space. An **inner product** is a function from $X \times X \rightarrow \mathbb{C}$

$$(x, y) \mapsto \langle x, y \rangle \quad (95)$$

if it satisfies, $\forall x, y, z \in X$

1. $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$
2. $\langle x, y \rangle = \overline{\langle y, x \rangle}$
3. $\langle x, x \rangle \in (0, \infty)$

The pair $(X, \langle \cdot, \cdot \rangle)$ is a **pre-Hilbert space**. It is also a Banach Space with norm induced by the inner product:

$$\| \cdot \| = \langle \cdot, \cdot \rangle^{1/2} \quad (96)$$

If X is also complete, it is a **Hilbert Space**

The Riez Representation Theorem

Theorem 2.3 (Riez Representation Theorem)

Let $f \in H^*$, $\exists! y \in H$ such that $\forall x \in H$ we have:

$$f(x) = \langle x, y \rangle \quad (97)$$

The Basis Theorem

Theorem 2.4 (Bessel's Inequality)

If $\{u_\alpha\}_{\alpha \in A}$ is an orthonormal set in H , then for any $x \in H$:

$$\sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2 \quad (98)$$

Note this implies $\{\alpha : \langle x, u_\alpha \rangle \neq 0\}$ is countable

Theorem 2.5 (Basis Theorem)

If $\{u_\alpha\}_{\alpha \in A}$ is an orthonormal set then the following are equivalent:

1. (*Completeness*) If $\langle x, u_\alpha \rangle = 0, \forall \alpha \in A$ then $x = 0$
2. (*Parseval's Identity*) $\forall x \in H$

$$\|x\|^2 = \sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2 \quad (99)$$

3. (*Representation*) $\forall x \in H$

$$x = \sum_{\alpha \in A} \langle x, u_\alpha \rangle u_\alpha \quad (100)$$

Proof 2.3

(1) $\implies$ (3) Let $x \in H$, let $\{\alpha_j\}$ be a enumeration of nonzero terms in the sum $\sum_{\alpha \in A} \langle x, u_\alpha \rangle u_\alpha$

$$\sum_{j=1}^{\infty} \langle x, u_{\alpha_j} \rangle u_{\alpha_j} = \sum_{\alpha \in A} \langle x, u_\alpha \rangle u_\alpha \quad (101)$$

Using orthonormality we have^a:

$$\begin{aligned} \left\| \sum_n^m \langle x, u_{\alpha_j} \rangle u_{\alpha_j} \right\|^2 &= \left\langle \sum_n^m \langle x, u_{\alpha_j} \rangle u_{\alpha_j}, \sum_n^m \langle x, u_{\alpha_k} \rangle u_{\alpha_k} \right\rangle \\ &= \sum_n^m |\langle x, u_{\alpha_k} \rangle|^2 \end{aligned} \quad (104)$$

By Bessel's inequality, we know $\sum_j |\langle x, u_{\alpha_j} \rangle|^2$ converges. This implies^b the tail can be made arbitrarily small, i.e., $\forall \epsilon > 0$, \exists $N(\epsilon)$ such that $\forall N < n < m$ we have:

$$\sum_n^m |\langle x, u_{\alpha_k} \rangle|^2 < \epsilon \quad (106)$$

^aThis is often referred to as the *Pythagorean's Theorem*, stating that for mutually orthogonal vectors $\{a_j\}_{j=1}^\infty$:

$$\left\| \sum_j a_j \right\|^2 = \sum_j \|a_j\|^2 \quad (102)$$

In particular, if we take only two vectors $a \perp b \in \mathbb{R}^2$, this becomes the Pythagorean's Theorem in middle school:

$$\|a\|^2 + \|b\|^2 = \|c\|^2 \quad (103)$$

^bSuppose we are confronted by the series $s = \sum_1^\infty a_j$ which converges absolutely to L . Need: given $\epsilon > 0$, there exists $N(\epsilon)$ such that $\forall N < n < m$ we have $\sum_n^m |a_j| < \epsilon$. We can pick N such that $\left| \sum_{j=1}^k |a_j| - L \right| < \epsilon/2, \forall k \geq N$. But now for all $m > n > N$, we have:

$$\sum_n^m |a_j| = \sum_1^m |a_j| - \sum_1^{n-1} |a_j| \quad (105)$$

The Sturm-Liouville Operator The **Sturm-Liouville operator** is a linear differential operator of the form

$$\mathcal{L} = -\frac{d}{dx} \left(p \frac{d}{dx} y \right) + q \frac{d}{dx} \quad (107)$$

We've shown that the Sturm-Liouville operator is self-adjoint with appropriate boundary condition in the sense that:

$$\langle u, \mathcal{L}v \rangle = \langle \mathcal{L}^*u, v \rangle \quad (108)$$

hence it is analogous to self-adjoint operator in finite-dimensional vector spaces, which as symmetric matrix representation when the underlying field is real. We now make this connection explicit, i.e., actually constructing the discretized, finite-dimensional matrix representation of the Sturm-Liouville Operator.

A Discrete Analog Let's consider a concrete SL operator defined on $[0, 1]$ with Dirichlet boundary condition $y(0) = y(1) = 0$. Instead of directly discretizing $\mathcal{L}$, we aim to find a matrix L such that discretizes the quadratic form:

$$F(y) = \int y \mathcal{L} y dx \rightarrow \hat{y}^T L \hat{y} \quad (109)$$

First we note that:

$$\begin{aligned}
 F(y) &= \int_0^1 y \left(-\frac{d}{dx} \left(p \frac{d}{dx} y \right) + qy \right) dx \\
 &= \int_0^1 -y \frac{d}{dx} \left(p \frac{d}{dx} y \right) dx + \int_0^1 qy^2 dx \\
 &= - \left\{ \left[y p \frac{dy}{dx} \right]_0^1 - \int_0^1 p \left(\frac{dy}{dx} \right)^2 \right\} + \int_0^1 qy^2 dx \\
 &= \int_0^1 p y'^2 + qy^2
 \end{aligned} \tag{110}$$

Now we discretize y, p, q to N -dimensional arrays:

$$\begin{aligned}
 \hat{y} &= [y_1, \dots, y_N] \\
 \hat{p} &= [p_1, \dots, p_N] \\
 \hat{q} &= [q_1, \dots, q_N]
 \end{aligned} \tag{111}$$

Using forward difference we get:

$$\hat{y} = \left(\frac{y_{i+1} - y_i}{h} \right)^2 \tag{112}$$

Rewrite the integral as a finite sum we obtain a discretized version of (110):

$$\hat{F}(\hat{y}) = \sum_{i=1}^N h p_i \left(\frac{y_{i+1} - y_i}{h} \right)^2 + q_i y_i^2 \tag{113}$$

Our goal is now to find a matrix L such that $\hat{y}^T L \hat{y} = \hat{F}(\hat{y})$. Rewrite $\hat{F}(\hat{y})$:

$$\hat{F}(\hat{y}) = \sum_{i=1}^N N p_i (y_{i+1}^2 - 2y_{i+1}y_i + y_i^2) + \frac{1}{N} q_i y_i^2 \tag{114}$$

We can split $L = L_1 + L_2$ such that

1. $\hat{y}^T L_1 \hat{y} = \sum_{i=1}^N N p_i (y_{i+1}^2 - 2y_{i+1}y_i + y_i^2)$
2. $\hat{y}^T L_2 \hat{y} = \sum_{i=1}^N \frac{1}{N} q_i y_i^2$.

(Constructing L_1) Let's first deal with L_1 . Consider the 2×2 block with the upperleft corner at (i, i) that should represent $N p_i (y_{i+1}^2 - 2y_{i+1}y_i + y_i^2)$. This is easy:

$$A_i = N p_i \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{115}$$

To assemble L_1 we can overlap A_i along the diagonal of L with a stride of 1. Concretely:

```

for i=1,...,N -1
    L[i,i] += Np_i
    L[i+1,i+1] += Np_i
    L[i,i+1] -= Np_i
    L[i+1,i] -= Np_i

```

Example 2.2

A quick example for $N = 4$:

$$4 \begin{pmatrix} p_1 & -p_1 & 0 & 0 \\ -p_1 & p_1 + p_2 & -p_2 & 0 \\ 0 & -p_2 & p_2 + p_3 & -p_3 \\ 0 & 0 & -p_3 & p_3 \end{pmatrix} \quad (116)$$

(Constructing L_2) L_2 is purely diagonal and is easier to derive. Since it only has quadratic term y_i^2 , at each row i we only need to include y_i . Hence:

$$L_2 = \frac{1}{N} \begin{pmatrix} q_1 & & \\ & \ddots & \\ & & q_N \end{pmatrix} \quad (117)$$

(Assembly $L_1 + L_2$) It is clear that L should look like:

$$L = N \begin{pmatrix} p_1 & -p_1 & & & \\ -p_1 & p_1 + p_2 & -p_2 & & \\ & \ddots & \ddots & \ddots & \\ & & -p_{N-1} & p_{N-1} + p_N & -p_N \\ & & & -p_N & p_N \end{pmatrix} + \frac{1}{N} \begin{pmatrix} q_1 & & & & \\ & \ddots & & & \\ & & & & q_N \end{pmatrix} \quad (118)$$

However, since we are dealing with concrete operators with accompanying domain and boundary condition, we need to encooperate the Dirchelt boundary condition $y(0) = 1, y(1) = 0$ into L . To do so we need to zero-out the first row and column of L . Hence finally we get:

Proposition 2.3

The Sturm-Liouville Operator on $[0,1]$ with Dirchlet boundary condition $y(0) = y(1)$ can be discretized as

$$L = N \begin{pmatrix} 0 & 0 & & & \\ 0 & p_1 + p_2 & -p_2 & & \\ & -p_2 & \ddots & \ddots & -p_{N-1} \\ & & \ddots & -p_{N-1} & p_{N-1} + p_N & 0 \\ & & & 0 & 0 & 0 \end{pmatrix} + \frac{1}{N} \begin{pmatrix} 0 & & & & \\ & q_2 & & & \\ & & \ddots & & \\ & & & & q_{N-1} \\ & & & & & 0 \end{pmatrix} \quad (119)$$

We can see that L is indeed a symmetric real matrix(Hermitian)!

Numpy Implementation This can easily be translated to naive numpy code:

```
class SturmLiouville:
    def __init__(self, N, p, q=None, grid='forward'):
        A = np.zeros(shape=(N,N))
        if grid == 'forward':
            for i in range(N-1):
                A[i, i] += p[i]
                A[i+1, i+1] += p[i]
                A[i, i+1] -= p[i]
                A[i+1, i] -= p[i]

        if q is not None:
            I = np.eye(N)
```

```

        L = A + q @ I
    else:
        L = A

    # Dirchlet Boundary Condition
    L[0,0]=0.0
    L[0,1]=0.0
    L[1,0]=0.0
    L[N -1,N -1]=0.0
    L[N -1,N -2]=0.0
    L[N -2,N -1]=0.0
    self.L = (L * N ** 2)
    self.N = N

def solve_eigen(self, top_k=10):
    eval, evec = np.linalg.eig(self.L)
    top_k_idx = np.argsort(eval[eval>0])[:top_k]
    return eval[top_k_idx] , evec[:, top_k_idx], top_k_idx

```

Example: $-\frac{d^2}{dx^2}$ as a Matrix We can do a quick example to showcase that L is indeed a discretization of $\mathcal{L}$.

Example 2.3 ($-\frac{d^2}{dx^2}$)

We have already seen that the operator:

$$T = -\frac{d^2}{dx^2}, D(T) = \{y, Ty \in L^2[0, 1] : y(0) = y(1) = 0\} \quad (120)$$

has eigenvalues and eigenvectors:

$$\begin{aligned} y_j(x) &= \sqrt{2} \sin j\pi x \\ \lambda_j &= j^2\pi^2 \end{aligned} \quad (121)$$

for $j = 1, 2, 3, \dots$. The problem is that, can we solve a matrix eigenvalue problem to recover exactly the same result? Indeed we can! Passing $p = -1$ and $q = 0$ gets us:

$$L = N \begin{pmatrix} 0 & 0 & & & \\ 0 & 2 & -1 & & \\ & -1 & \ddots & \ddots & \\ & & \ddots & 2 & -1 \\ & & & -1 & 2 & 0 \\ & & & & 0 & 0 \end{pmatrix} \quad (122)$$

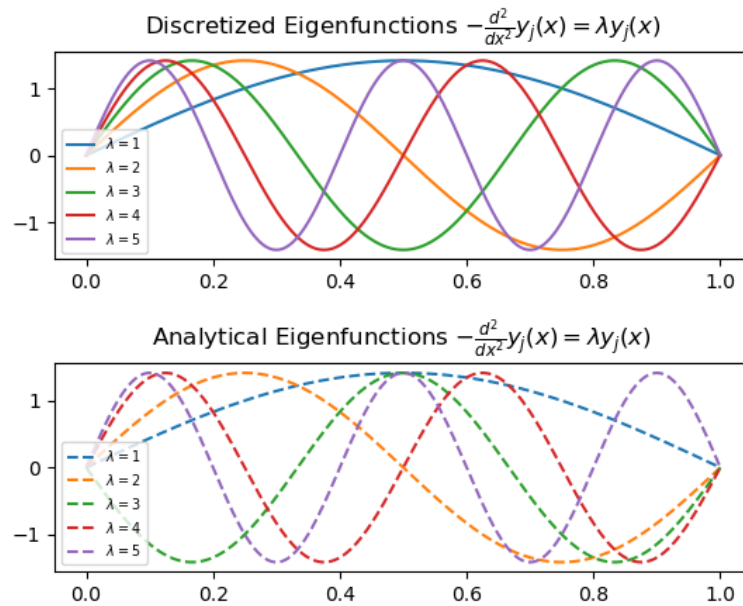
and solving the matrix eigenvalue problem

$$L\hat{y} = \lambda\hat{y} \quad (123)$$

gets us: ^a

^aSince iterative eigensolvers gives eigenvectors of arbitrary magnitude $v_j = a \sin(j\pi(x))$, we need to normalize properly by 1. Ensure $a > 0$

1. Divide by $\sqrt{\sum y_i^2 h}$



Exercise 2.2

1. We discretized L based on the quadratic form $\int_0^1 y \mathcal{L} y dx$, hence $\hat{y}^T L \hat{y}$ should coincide with $\int_0^1 y \mathcal{L} y dx$. Come up with at least 2 functions and 2 operators that satisfy the requirement ($L^2(0,1), y(0) = y(1) = 0$). Compute the quadratic form analytically and numerically. Check that the results are the same.
2. Remove the masking code snippet (which implements Dirichlet boundary condition) in the naive SL class `SturmLiouville`. What is the resulting eigenvectors? Explain the result.
3. Try coming up with different discretization with center difference or back difference. Implement it in `python` or `julia`.
 - Is the resulting matrix L still symmetric?
 - Center differencing scheme results in a numerical artifact, what is the numerical artifact? Try explaining why such artifact occur (*Hint: Note that even and odd indices are decoupled*)

CHAPTER III
Calculus of Variation

3.1 Euler-Lagrange Equation

Differential Equation from Physics

A lot of physical science and engineering is about finding a reduction from a physical problem to differential equations. The newtownian way is to zoom into some local point of the system and derive a set of constraint that holds in the neighborhood. In the case of the vibration of a string, for example, we would find some local segment of string and analyze the tension that acts on it:

$$T(\sin(\theta_1) - \sin(\theta_2)) = m \frac{\partial^2 u}{\partial t^2} \quad (124)$$

where u is the vertical displacement from equilibrium. We then take the calculus limit and find out that locally the acceleation of the vertical displacement should be proportional to the curvature. Hence the one-dimensional wave equation:

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} \quad (125)$$

1

The Lagrangian way is the opposite, we recognize some global objective the system need to be optimized toward, i.e., minimizing the lagragian $L = T - V$ where T is *kinetic energy* and V is *potential energy*. Then, by introduce some perturbation, we extract a constraint which also manifest itself as a differential equation. The following sections descirbes this process in detail.

Newtonian and Lagrangian Mechanics

- Newtonian mechanics compute the balance of forces locally and obtain differential equation through

$$F = m \frac{d^2 \mathbf{x}}{dt^2} \quad (126)$$

- Lagrangian mechanics translate the physics into some global optimization problem and reduce to differential equation through the Euler-Lagrange equation

$$\frac{\partial L}{\partial y} - \frac{d}{dx} \frac{\partial L}{\partial y'} = 0 \quad (127)$$

Functional : A global optimization objective

A functional in *calculus of variation* is a map from a function to a real number, i.e., $J : C^\infty \rightarrow \mathbb{R}$. In particular, it mostly comes in some sort of a integral. The simplest form being analgous to that of the lagrangian:

$$J[y] = \int_{x_1}^{x_2} f(x, y, y') dx \quad (128)$$

The above example suffice to establish the explict connection of (128)to Lagrangian:

Example 3.1 (The lagrangian for the vibrating string)

Let u be deviation from the equilibrium position

1. The kinetic energy:

$$T = \frac{1}{2} \int_0^L \rho u_t(x, t)^2 dx \quad (129)$$

¹Here a is a physical parameter intrinsic to the string, i.e., $\sqrt{T/\rho}$ where T is the tension at equilibrium adn ρ is the density.

where L is the length of the string

2. The potential energy:

$$V = \frac{1}{2}k \int_0^L (1 + u_x^2) dx \quad (130)$$

a

The lagrangian is hence:

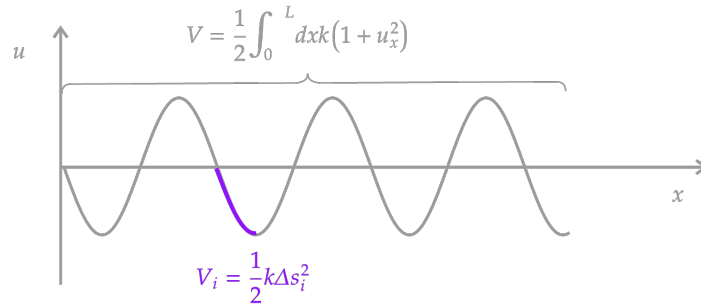
$$L = \int_0^L \frac{1}{2} \left(\rho u_t(x, t)^2 - k(1 + u_x^2) \right) dx \quad (133)$$

^aIt is tempting for some of us who are not trained as a physics student to write the potential energy as $\frac{1}{2} \left(\int_a^b \sqrt{1 + u_x^2} dx \right)^2$. The rationale is as follows: if we pause at a moment and straighten the vibrating string as a line. Then we observe how much elongation had happened and use hooke's law on the entire string. This is physically wrong since the elongation happens inhomogenously hence we have to apply hooke's law locally and integrate. Also mathematically,

$$\left(\int f \right)^2 = \int f^2 \quad (131)$$

iff f is some constant function. The general relationship can be obtained from Cauchy-Schwarz inequality by taking $f = \sqrt{1 + u_x^2}$ and $g = 1$

$$\left(\int_0^L \sqrt{1 + u_x^2} dx \right)^2 \leq \int_0^L (1 + u_x^2) dx \quad (132)$$



The potential energy of a vibrating string.

We now scrutinize how exactly we optimize (128).

Euler-Lagrange Equation

Theorem 3.1 (Euler-Lagrange Equation)

y is a stationary point for

$$J[y] = \int_{x_1}^{x_2} f(x, y, y^{(1)}, \dots, y^{(n)}) dx \quad (134)$$

iff:

$$\boxed{\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0} \quad (135)$$

The boxed equation is known as the **Euler-Lagrange Equation**

Discretized Euler-Lagrange Equation**First Integral****Definition 3.1 (First Integral)**

Suppose

$$y^{(n)} = f(x, y, y^{(1)}, \dots, y^{(n-1)}) \quad (136)$$

is an ordinary differential equation of order n . We can define an equivalent n -dimensional system by letting:

$$\begin{aligned} y_1 &= y \\ y_2 &= y^{(1)} \\ &\vdots \\ y_n &= y^{(n-1)} \end{aligned} \quad (137)$$

Denote by:

$$\frac{dy_i}{dx} = f_i(x, y_1, \dots, y_n) \quad (138)$$

Now suppose f is:

1. Continuous about $x, y_1, \dots, y_n$
2. Continuously differentiable about $y_1, \dots, y_n$

Define the *first integral* of the n^{th} -order problem to be a function $I = I(x, y_1, \dots, y_n)$ that also satisfies the above two requirements, and furthermore has the definitive property:

$$I(x, y_1, \dots, y_n) = \text{Constant} \quad (139)$$

along solution curves $\Gamma = (y_1(x), y_2(x), \dots, y_n(x))$ with the constant determined by the solution curve.

We are confronted by the higher-order ordinary differential equation:

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0 \quad (140)$$

The compute the first integral of this equation is to rewrite the above to obtain an equation of the form:

$$\frac{d}{dt} I(x, y, y', \dots) = 0 \quad (141)$$

The strategy is to compute the difference between (140) and the exact differential:

$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \sum_{j \geq 0} \frac{\partial f}{\partial y^{(j)}} \frac{dy^{(j)}}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial y^{(1)}} \frac{dy^{(1)}}{dx} + \dots \quad (142)$$

If we multiply (140) by y' and subtract the two equations we get:

$$\frac{df}{dx} - y' \left\{ \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right\} = \frac{\partial f}{\partial x} + y'' \frac{\partial f}{\partial y'} + y' \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) - \sum_{j \geq 2} \frac{\partial f}{\partial y^{(j)}} \frac{dy^{(j)}}{dx} \quad (143)$$

The particular form that allows easy first integral

If we assume f is only dependent on y, y' , the above becomes $f(y, y')$, then

$$\begin{aligned} \sum_{j \geq 2} \frac{\partial f}{\partial y^{(j)}} \frac{dy^{(j)}}{dx} &= 0 \\ \frac{\partial f}{\partial x} &= 0 \end{aligned} \quad (144)$$

Now:

$$\frac{df}{dx} - y' \underbrace{\left\{ \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right\}}_0 = \frac{d}{dx} \left(y' \frac{\partial f}{\partial y'} \right) \quad (145)$$

Hence:

$$\frac{d}{dx} \left(f - y' \frac{\partial f}{\partial y'} \right) = 0 \quad (146)$$

We have successfully obtained the first integral:

$$I(y, y) = f - y' \frac{\partial f}{\partial y'} \quad (147)$$

Exercises

Euler-Lagrange Equation for Multiple Independent Variables Prove that:

Theorem 3.2 (Euler-Lagrange Equation)

Consider the functional

$$J[y] = \int f(x_1, x_2, \dots, x_n, y_{x_1}, y_{x_2}, \dots, y_{x_n}) d\mathbf{x} \quad (148)$$

Show that the stationary is achieved iff:

$$\frac{\partial f}{\partial y} - \sum_i \frac{d}{dx} \left(\frac{\partial f}{\partial y_{x_i}} \right) \quad (149)$$

Fermat's Principle

Note

^aFermat's Principle

The path taken by a ray of light between any two point makes stationary the travel time between those points.

^aFrom Dr. P.Draper's homework problems for PHYS 508 (*Mathematical Physics I*) at UIUC.

We also define:

Definition 3.2 (Refractive Index)

A medium is characterized by the speed of light traveling through. Specifically, the refractive index is defined by $n = \frac{c}{\text{speed of light traveling in the medium}}$

Problem 1 Light Propagates in Straight Line in Homogenous Media

1. Show that light propagates along a straight line in homogenous media.

Solution

Since n is independent of position in homogenous media, make stationary the travel time is equivalent to make stationary the length of the path taken. Now the problem is equivalent to optimizing:

$$\int_{x_1}^{x_2} \sqrt{1 + y'^2} dx \quad (150)$$

Note:

$$f_y = 0 \quad f'_y = \frac{y'}{\sqrt{1 + y'^2}} \quad (151)$$

The Euler-Lagrange equation is, therefore:

$$\begin{aligned} \frac{d}{dx} \left[\frac{y'}{\sqrt{1 + y'^2}} \right] &= 0 \\ \frac{y'}{\sqrt{1 + y'^2}} &= C_1 \end{aligned} \quad (152)$$

Note the L.H.S is a non-constant function of y' , hence:

$$y' = C_2 \implies y = C_2 x + C_3 \quad (153)$$

Problem 2 Snell's Law

Consider the propagation of light from a flat slab of glass of refractive index n_1 into to another with refractive index n_2 . By examining paths that need not be differentiable at the flat interface, establish Snell's law:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2) \quad (154)$$

Solution

We consider the first point at $(x_1, 0)$, the second point at (x_2, y_2) where $x_1 < 0 < x_2$. The time taken is hence:

$$\int_{x_1}^{x_2} \frac{1}{v(x)} \sqrt{1 + y'^2} dx \quad (155)$$

where:

$$v(x) = \begin{cases} c/n_1 & \text{if } x < 0 \\ c/n_2 & \text{if } x > 0 \end{cases} \quad (156)$$

The Euler-Lagrange equation is:

$$0 + \frac{d}{dx} \left(\frac{y'}{v(x) \sqrt{1 + y'^2}} \right) = 0 \quad (157)$$

Hence:

$$\frac{y'}{v(x) \sqrt{1 + y'^2}} = C \quad (158)$$

For $x < 0$, we have:

$$\frac{1}{v(x)} \frac{y'}{\sqrt{1+y'^2}} = \frac{n_1}{c} \sin(\theta_1) = C_1 \quad (159)$$

For $x > 0$, we have:

$$\frac{1}{v(x)} \frac{y'}{\sqrt{1+y'^2}} = \frac{n_2}{c} \sin(\theta_2) = C_2 \quad (160)$$

Hence:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2) \quad (161)$$

Question: If we flip the axis we would get $n_1 \cos \theta_1 = n_2 \cos \theta_2$, why is that wrong?

Problem 3 General Snell's law

A planar light ray propagates in an layered medium with refractive index $n(x)$. Use Fermat's principle to establish a generalized Snell's law:

$$n(x) \sin(\psi(x)) = \text{constant} \quad (162)$$

by finding the equation for stationary paths for

$$F_1 = \int n(x) \sqrt{1+y'^2} dx \quad (163)$$

(Here the prime denotes differentiation with respect to x .) Repeat this exercise with x, y swapped by finding a similar equation for the stationary paths of

$$F_2 = \int n(y) \sqrt{1+y'^2} dx \quad (164)$$

By using suitable definitions of the angle of incidence ψ , in each case show that the two formulations of the problem of a layered medium give physically equivalent answers. (In the second formulation you will find it easiest to use the first integral.)

Drums and Membranes ²The shape of a distorted drumskin is described by the function $h(x, y)$, which gives the height to which the point (x, y) of the flat undistorted drumskin is displaced.

Problem 1 Area of distorted drumskin

Show that the area of the distorted drumskin is equal to

$$A[h] = \int dx dy \sqrt{1 + \partial_x h^2 + \partial_y h^2} \quad (165)$$

Solution

Consider the area differential $dA = ds_x ds_y$ where $ds_x = \sqrt{1 + \partial_x h^2} dx$ and $ds_y = \sqrt{1 + \partial_y h^2} dy$ hence

$$dA = \sqrt{1 + \partial_x h^2 + \partial_y h^2 + (\partial_x h \partial_y h)^2} dx dy \quad (166)$$

since $\partial_x h \partial_y h \in \mathcal{O}(d\rho^2)$ where $d\rho = \sqrt{dx^2 + dy^2}$ we can drop it. As desired.

²From Dr. P.Draper's homework problems for PHYS 508(*Mathematical Physics I*) at UIUC.

Area for small distortion

Show that for small distortions, the area reduces to

$$A[h] = \frac{1}{2} \int dx dy |\nabla h|^2 + \text{constant} \quad (167)$$

Solution

Recall the generalized binomial theorem:

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots + \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!} x^n + \dots \quad (168)$$

for $|x| < 1$. As a result:

$$(1+x)^{1/2} = 1 + \frac{1}{2}x + \mathcal{O}(x^2) \quad (169)$$

Assume a small perturbation, in particular, assume $|\nabla h|^2 \ll 1$, then

$$\sqrt{1+|\nabla h|^2} = 1 + \frac{1}{2}|\nabla h|^2 + \mathcal{O}(|\nabla h|^4) \quad (170)$$

As a result,

$$A[h] = \frac{1}{2} \int dx dy |\nabla h|^2 + \underbrace{\int dx dy}_{\text{constant}} \quad (171)$$

Problem 3

Show that if h satisfies the two-dimensional Laplace equation then A is stationary with respect to variations that vanish at the boundary.

Solution

We need that the Euler-Lagrange equation for (167) is equivalent to the two-dimensional Laplace equation. That is simple: let $f(x, y, h, h_x, h_y) = h_x^2 + h_y^2$,

$$\begin{aligned} \frac{\partial f}{\partial h} + \frac{d}{dx} \left(\frac{\partial f}{\partial h_x} \right) + \frac{d}{dy} \left(\frac{\partial f}{\partial h_y} \right) &= 0 \\ 0 + 2 \left(\frac{dh_x}{dx} + \frac{dh_y}{dy} \right) &= 0 \\ \nabla^2 h &= 0 \end{aligned} \quad (172)$$

Problem 4

Suppose the drumskin has mass ρ_0 per unit area, and surface tension T . Write down the Lagrangian controlling the motion of the drumskin and derive the equation of motion that follows from it.

Solution

The kinetic term is given by:

$$T = \frac{1}{2} \int_{\Omega} dx dy \rho_0 h_t^2 \quad (173)$$

The potential term is: ^a

$$V = \int_{\Omega} dx dy T |\nabla h|^2 \quad (174)$$

Hence the lagrangian is:

$$\begin{aligned} L &= T - V \\ &= \int_{\Omega} dx dy \left\{ \frac{1}{2} \rho_0 h_t^2 - T |\nabla h|^2 \right\} \end{aligned} \quad (175)$$

Computing the various derivatives:

$$\frac{\partial L}{\partial h} = 0 \quad \frac{\partial L}{\partial h_t} = \rho_0 h_t \quad \frac{\partial L}{\partial h_x} = -2Th_x \quad \frac{\partial L}{\partial h_y} = -2Th_y \quad (176)$$

^aThe key to solve this problem is actually to realize that surface tension has the dimension $[F]/[L] = [E]/[L]^2$, in otherword, surface tension is precisely the perunit potential energy due to “stretching”/“distorsion” by construction.

3.2 Generalizations

This section focuses on more general cases. In particular,

1. **Multiple Constraint**, i.e., in the case of

$$J = \int f(x, y_1, y_2, \dots, y_n, \dot{y}_1, \dot{y}_2, \dots, \dot{y}_n, \ddot{y}_1, \dots) \quad (177)$$

2. **Variable Endpoints** In previous cases, we used the key assumption that:

Fixed Endpoint Assumption

The perturbation δy is 0 at endpoints, hence all boundary/surface terms arising from integration by parts may be ignored.

We now relax this assumption but rather work out boundary conditions based on the stationary condition with variable endpoints. This assumption can in turn be applied to the resulting differential equations.

3. **Constrained Optimization** Often, the minimization of the action is subjected to physical constraint. For instance, in the famous *catenary problem*, we need to fix the length of the hanging chain.

Multiple Constraint

Variable Endpoints

Constraint Optimization

Exercises

Elastic Rods

Elastic Rods

- The **elastic energy per unit length** of a bent steel rod is given by

$$\frac{1}{2} \frac{YI}{R^2} \quad (178)$$

Here Y is the *Young's modulus* of the steel and I is the moment of inertia of the rod's cross section about an axis through its centroid and perpendicular to the plane in which the rod is bent.

- Let us assume the rod is only slightly bent into the yz plane and lies close to the z axis. R is the radius of curvature due to the bending, related to the curvature by

$$\frac{1}{R} = \kappa = |T_z| \quad (179)$$

where T is the unit tangent vector to the rod.

Problem 0 Show that the elastic energy can be approximated by:

$$U[y] = \int_0^L \frac{1}{2} \left\{ \frac{1}{2} YI (y'')^2 \right\} dz \quad (180)$$

Problem 1 The rod is used as a column which supports a compressive load Mg directed along the z axis (which is vertical). Show that :

(a) when the rod buckles slightly (i.e. bends with both ends remaining on the z axis) the total energy, including the gravitational potential energy of the loading mass M , can be approximated

by

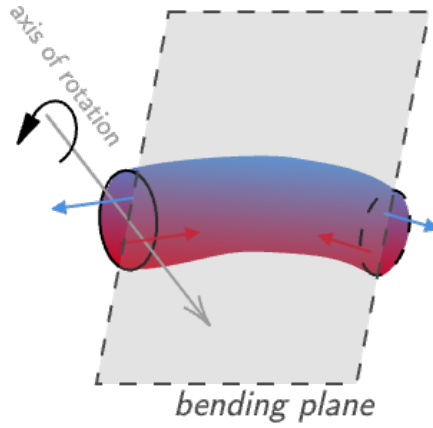
$$U[y] = \int_0^L \left\{ \frac{YL}{2} (y'')^2 - \frac{Mg}{2} (y')^2 \right\} dz \quad (181)$$

(b) By considering deformations of the form

$$y(z) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi z}{L}\right) \quad (182)$$

show that the column is unstable to buckling and collapse once

$$Mg \geq \frac{\pi^2}{L^2} YI \quad (183)$$



Bending beam and the axis of rotation associated with the moment of inertia I

Solution to Problem 0

We need a little lemma(or recall from Calculus I):

$$\kappa = \frac{y''}{(1 + y'^2)^{3/2}} \quad (184)$$

For small perturbation $|y'| \ll 1$ hence $(1 + y'^2)^{3/2} \approx 1$ hence $\kappa = \frac{1}{R} = (y'')$. Squaring it we have :

$$\frac{1}{R^2} = (y'')^2 \quad (185)$$

Hence:

$$dU = \frac{1}{2} YI (y'')^2 dz \quad (186)$$

Integrating get us the desired result.

Proof 3.1 (Proof of lemma)

If we imagine we traverse along a curve Γ , the acceleration would be:

$$\begin{aligned} r''(z) &= \frac{d}{dz} \left(\frac{dr}{dz} \right) \\ &= \frac{d}{dz} \left(\frac{ds}{dz} T \right) \\ &= \frac{d^2s}{dz^2} T + \frac{ds}{dz} \frac{dT}{dz} \end{aligned} \quad (187)$$

Note that

$$\left\langle T, \frac{dT}{dz} \right\rangle = 0 \quad (188)$$

since $|T| = 1$. Therefore $r''(z)$ can be decomposed to a tangential component $\frac{d^2s}{dz^2} T$ and a centrepetal component $\frac{ds}{dz} \frac{dT}{dz}$. Also, $\frac{dT}{dz}$ is co-linear with our quantity of interest

$$\frac{dT}{dz} = \frac{dT}{ds} \frac{ds}{dz} \quad (189)$$

Therefore:

$$r''(z) = \frac{d^2s}{dz^2} T + \left(\frac{ds}{dz} \right)^2 \frac{dT}{ds} \quad (190)$$

We can isolate the centrepetal component by apply a cross product:

$$T \times r''(z) = T \times \left(\frac{ds}{dz} \right)^2 \frac{dT}{ds} \quad (191)$$

Now :

1. (LHS) $T = (1, 0, y')/ds$, hence $r''(z) = (0, 0, y'')$. Hence the cross product is:

$$\frac{1}{\sqrt{1+y'^2}} (1, 0, y')^T \times (0, 0, y'') = \frac{1}{\sqrt{1+y'^2}} (0, y'', 0)^T \quad (192)$$

2. (RHS) Since $T, \frac{dT}{dz}$ are orthogonal to each other and both lie in the x-z plane, we have:

$$T \times \left(\frac{ds}{dz} \right)^2 \frac{dT}{dz} = \left(\frac{ds}{dz} \right)^2 \left| \frac{dT}{dz} \right| (0, 1, 0)^T \quad (193)$$

Now combining the above two equations, we get:

$$\frac{y''}{\sqrt{1+y'^2}} = \left(\frac{ds}{dz} \right)^2 \left| \frac{dT}{dz} \right| \quad (194)$$

As desired.

Solution to Problem 1

(a) Consider the schemetic above. Suppose $y = 0$ for $z \in (L - dL, L)$, then

$$\begin{aligned} L + dL &= \int_0^L \sqrt{1 + y'^2} dz \\ &\approx \int_0^L 1 + \frac{1}{2} y'^2 dz \\ &= L + \int_0^L \frac{1}{2} y'^2 dz \end{aligned} \quad (195)$$

Hence $dL = \int_0^L \frac{1}{2} y'^2 dz$. If we consider the gravitational potential energy to be 0 at $h = L$, then the gravitational potential energy becomes:

$$- \int_0^L \frac{1}{2} y'^2 dz \quad (196)$$

the negative sign is there to account for compression. Combining with the result from problem 0 we get as desired.

(b) The Euler-Lagrange equation for the total energy functional is:

$$\begin{aligned} -\frac{d}{dx} \frac{\partial f}{\partial y'} + \frac{d^2}{dx^2} \frac{\partial f}{\partial y''} &= 0 \\ Mgy^{(2)} + YIy^{(4)} &= 0 \end{aligned} \quad (197)$$

Consider a series solution of the form:

$$y(z) = \sum_{n=1}^{\infty} a_n \sin(\omega_n z) \quad (198)$$

where $\omega_n = \frac{n\pi}{L}$. We have:

$$\begin{aligned} \frac{\widehat{\delta J}}{\delta f} &= \sum_{n=1}^{\infty} b_n \sin(\omega_n z) \\ &= \sum_{n=1}^{\infty} \underbrace{a_n (YI\omega_n^4 - Mgy_n^2)}_{b_n} \sin(\omega_n z) \end{aligned} \quad (199)$$

Hence a trivial solution is $a_n = 0, \forall n \geq 1$ which corresponds to a *stable, stiff rod*.

The rod buckles if, the trivial solution is no longer a local minimum, i.e., the hessian is no longer positive-definite. We need to find the hessian in this case.

Define the hessian in the spectral space as:

$$\hat{H} = \begin{pmatrix} \frac{\partial b_1}{\partial a_1} & \frac{\partial b_1}{\partial a_2} & \cdots \\ \frac{\partial b_2}{\partial a_1} & \frac{\partial b_2}{\partial a_2} & \cdots \\ \vdots & & \ddots \end{pmatrix} \quad (200)$$

which in our case is already diagonalized:

$$\hat{H} = \text{diag}(\{(YI\omega_n^4 - Mgy_n^2)\}_{n=1}^{\infty}) \quad (201)$$

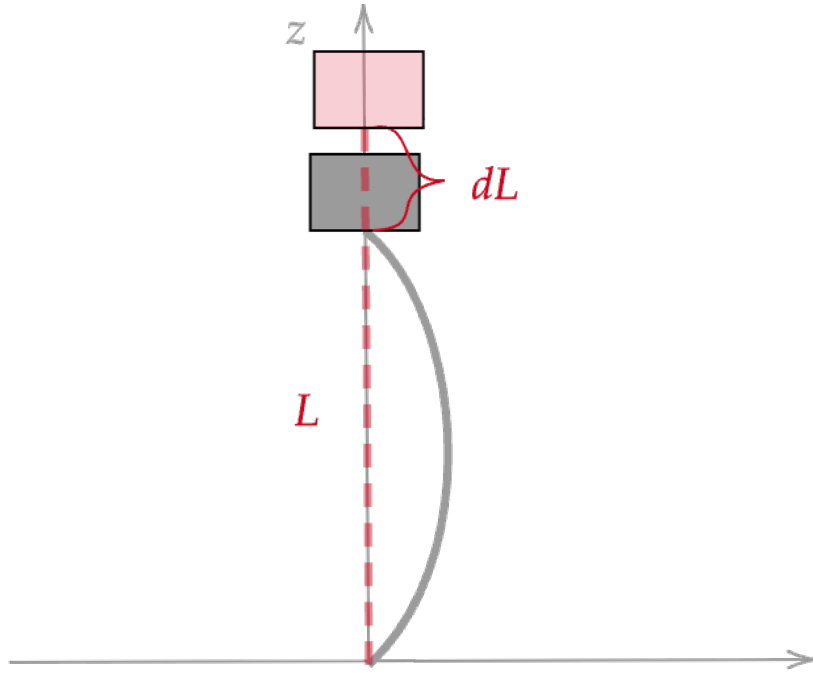
Local minimum requires positive-definiteness, which is :

$$\begin{aligned} YI\omega_n^4 - Mgy_n^2 &> 0 \\ M &< \frac{YI}{g} \omega_n^2 \end{aligned} \quad (202)$$

Picking $n = 1$ (smallest eigenvalue) satisfies the last inequality.

$$M < \frac{Y I \pi^2}{g L^2} \quad (203)$$

As desired.



CHAPTER IV
Partial Differential Euqations

4.1 Introduction

Definition**Definition 4.1**

A PDE for a function $u(x_1, \dots, x_n)$ is of the form:

$$F(x_1, \dots, x_n, u, u_{x_1}, \dots, u_{x_n}, u_{x_1 x_1}, u_{x_1 x_2}, \dots) = 0 \quad (204)$$

1. The *order* is the order of the highest-order derivative
2. An equation is called *linear* if it is linear on u and its derivatives
3. An equation is called *semi-linear* if all coefficients are only functions of $x_1, \dots, x_n$
4. An equation is called *quasilinear* if coefficient a_N is only functions of x, u and lower-order derivatives.

4.2 Elementary Methods

Conservation Laws

The Method of Characteristics

Integral Surface Consider the general quasilinear first order PDE:

$$a(x, y, u)u_x + b(x, y, u)u_y = c(x, y, u) \quad (205)$$

We note that the vector field $(-u_x, -u_y, 1)^T$ would be orthogonal to the solution $u(x, y)$ everywhere. Therefore the vector field $(a, b, c)^T$ is perpendicular to $(-u_x, -u_y, 1)^T$ thus must lie in the tangent plane of $u(x, y)$. Surfaces that are tangent to a vector space is called *integral surfaces*, hence to find the solution we want to find the integral surfaces of $u(x, y)$.

We do this by:

Method of Characterisitcs

1. Parametrize the inital condition $\Gamma(r)$
2. Find integral curves χ starting from one instance of the initial condition(at some r) by solving a system of ODE. Concretely, suppose $\chi(s) = (x(s), y(s), z(s))$ is a parametrization of the integral curves, then it must satisfy that:

$$\begin{cases} \frac{dx}{ds} = a(x, y, u) \\ \frac{dy}{ds} = b(x, y, u) \\ \frac{dz}{ds} = c(x, y, z) \end{cases} \quad (206)$$

on some small $|t - t_0| < \delta$. The resulting integral curves are functions of r, s where r specifies where the evolution start and s specifies how much the integral curve grow away from the Γ .

3. Compute the integral surface as a continuous union by replacing r, s with x, y

Example 4.1 (Constant Wave-speed Advection)

Consider the equation $u_t + cu_x = 0$ with the initial condition $u(t, 0) = \exp(-x^2)$. We parametrize the initial condition as $(t, x, z) = (0, r, \exp(-r^2))$. Fix r , the system is:

$$\begin{cases} \frac{dt}{ds} = 1 \\ \frac{dx}{ds} = c \\ \frac{dz}{ds} = 0 \end{cases} \implies \begin{cases} t(r, s) = s + C(r) \\ x(r, s) = cs + D(r) \\ z(r, s) = E(r) \end{cases} \quad (207)$$

Substituting the initial condition, we get:

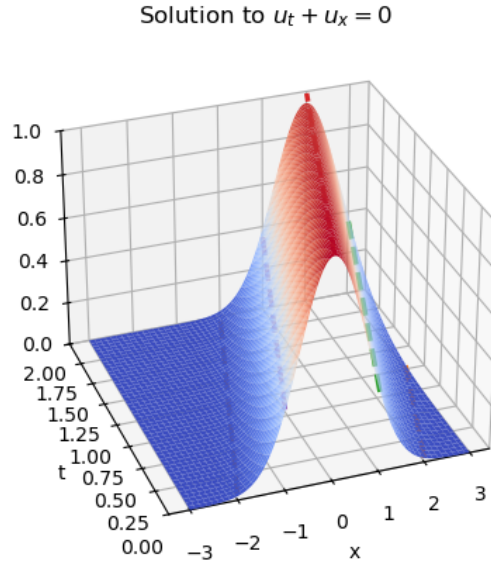
$$\begin{cases} t(r, s) = s \\ x(r, s) = cs + r \\ z(r, s) = \exp(-r^2) \end{cases} \quad (208)$$

But now $r = x - cs = x - ct$ hence we get the solution:

$$u(t, x) = \exp(-(x - ct)^2) \quad (209)$$

Visualizing the solution:

The Finite Volume Method



We showed the special case when $c = 1$, the lines are the integral curves above. In this special case the value of $z = u(x, y)$ is fixed on the curve.

Hyperbolic Conservation Law For simplicity, we consider the homogenous case:

$$u_t + a(t, x)u_x = 0 \quad (210)$$

If we discretize the spatial domain into an array of contiguous cells called *control volumes*, we can see that the equation encapsulate a sense of *conservation*. Suppose $[a, b]$ is a cell of interest, integrate the equation by parts results in :

$$\frac{d}{dt} \int_a^b u dx = \underbrace{a(t, a)u(t, a)}_{\text{flux in}} - \underbrace{a(t, b)u(t, b)}_{\text{flux out}} - \underbrace{\int_a^b a_x(t, x)u dx}_{\text{source/dissipation}} \quad (211)$$

In particular, if we have $a(t, x) = a(t)$, then we have no source and dissipation. This motivates us to define:

Definition 4.2 (Hyperbolic Conservation Law)

There are two subclasses, **strict conservation law**(with no source term) and **balance law**(conservation law with source term).

1. Conservation Law

$$u_t + (f(u))_x = 0 \quad (212)$$

2. Balance Law

$$u_t + (f(u))_x = g(u, x, t) \quad (213)$$

For a simple strict conservation law we have:

$$\frac{d}{dt} \int_a^b u = \underbrace{f(u(a, t))}_{\text{flux in}} - \underbrace{f(u(b, t))}_{\text{flux out}} \quad (214)$$

It is clear that:

Proposition 4.1

The general homogenous transport equation $u_t + a(x, t)u_x = 0$ is a strict conservation law iff $a(x, t) = a(t)$

The idea behind finite volume

Discretization

Finite volume discretize the space into cells $x_i, i \in \mathbb{N}$ being the cell center and $x_{i-1/2}, x_{i+1/2}$ being the faces of the cell.

Finite volume method utilise the conservation structure of the equation. Specifically, we enforce that conservation law holds at each cell $[x_{i-1/2}, x_{i+1/2}]$

$$\int_{t_j}^{t_{j+1}} \left\{ \int_{x_{i-1/2}}^{x_{i+1/2}} [u_t + (f(u))_x - g] dx \right\} dt = 0 \quad (215)$$

Wave Equation

The d -dimensional wave equation is given by the form:

$$u_{tt} = a^2 \nabla^2 u + f(\mathbf{x}, t) \quad (216)$$

where $u(\mathbf{x}, t) : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$.

The one dimensional wave equation $u_{tt} = a^2 u_{xx}$ very naturally arise in modeling of a *vibrating string*.

The Vibrating String Consider a horizontally placed string with length L . We make the following assumptions

Assumptions

1. Suppose at equilibrium the tension is a constant T . As the string vibrate at small amplitude, we can assume at arbitrary position the tension is a constant and takes the value of the equilibrium tension T .
2. The string vibrates in the vertical direction, i.e., no horizontal stretching
3. Let $\rho(x)$ be the density distribution of the string.

Now to model the vibration mathematically, **we place the string along the x -axis, let $u(x, t)$ be the vertical displacement of the string at position x from equilibrium position at time t .**

The Lagrangian Way It's clear that the kinetic energy of the string should be given by:

$$K = \frac{1}{2} \int_0^L m u_t^2 dx \quad (217)$$

Poisson Equation

Interactive Visualization !

This notebook contain interactive visualization that needs binder connection. Please click the **power button on the top right corner** before you start reading ! Establishing the connection and building the container takes a while, so clicking it before you read would make sure the interactive code to be ready to run when you scroll on to them.

2D Poisson Equation Let $\Omega \subset \mathbb{R}^2$, we consider the Poisson equation with homogenous boundary condition:

Definition 4.3 (2D Poisson Equation with Homogenous Boundary Condition)

$$\begin{aligned} -\Delta u &= f(x, y) & (x, y) \in \Omega \\ u(x, y) &= 0 & (x, y) \in \Gamma_1 \\ \frac{\partial u}{\partial \nu}(x, y) &= 0 & (x, y) \in \Gamma_2 \end{aligned} \quad (218)$$

where $\Gamma_1 \cup \Gamma_2 = \partial\Omega$ such that both Γ_1, Γ_2 is measure-zero. ν is the outward normal vector.

Spectral Method Let

$$u(x, y) = \sum_{-\infty < p, q < \infty} a_{pq} \exp(i(px + qy)) \quad (219)$$

Which has:

$$\Delta u = \sum_{-\infty < p, q < \infty} a_{pq} \quad (220)$$

Also let the fourier transform of $f(x, y)$ be $\mathcal{F}(f(x, y)) = F(p, q)$.

Finite-Element Method(FEM)

The Tent Function We consider the

Viualizing the Tent Basis

Tip

If you have not do so, click the power button on the top right corner. After everything builds up, you can click the start button . You should see two sliders showing up below.

- The resolution slider determines how “dense” these tent basis are in filling the domain. Lower resolution means larger and less tents and vice versa.
- The position slider let you adjust which particular basis function you are watching. In otherword, where the “tent” is located in the domain Ω

We can show that:

Proposition 4.2

Suppose $u \in C^2(\bar{\Omega})$ is a solution of (218), then $\forall v \in H_{\Gamma_1}^1(\Omega)$, we have:

$$\int_{\Omega} \nabla u \nabla v = \int_{\Omega} f v \quad (221)$$

Proof 4.1

We first show *Green's Formula*(change of variable):

$$\int_{\Omega} v \Delta u = \int_{\partial\Omega} v \frac{\partial u}{\partial \nu} - \int_{\Omega} \nabla u \nabla v \quad (222)$$

This motivates us to define a solution in a weaker sense. Since $a(u, v) = (f, v)$ is a necessary condition of u , how about making every u that satisfies this equation a “weak solution”(or generalized solution)?

We first define the space that our weak solution would reside:

$$H_{\Gamma_1}^1(\Omega) = \{f \in H^1(\Omega) : f = 0 \text{ on } \Gamma_1\} \quad (223)$$

CHAPTER V
Complex Variable

5.1 Analytical Function

Real Derivative and Complex Derivative

A complex function $f : \mathbb{C} \rightarrow \mathbb{C}$ resembles, more or less, functions like $\mathbb{R}^2 \rightarrow \mathbb{R}^2$. For instance, the complex exponential $f(z) = \exp(z)$ would be more or less similar to

$$f \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \exp(x) \cos(y) \\ \exp(x) \sin(y) \end{pmatrix} \quad (224)$$

Since the complex version can be rewritten as $\exp(z) = \exp(x + iy) = \exp(x)(\cos(y) + i \sin(y))$.

However, there does exist crucial differences. Recall for real functions $(\mathbb{R}^2, \mathbb{R}^2)$, they are differentiable if there exists a linear transformation $A \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ such that:

$$\lim_{x \rightarrow x_0} \frac{\|f(x_0 + h) - f(x_0) - Ah\|}{\|h\|} = 0 \quad (225)$$

If such A exists then it must equal to the **Jacobian** $Df(x_0)$ defined as:

$$\left(\frac{\partial f_i}{\partial x_j}(x_0) \right)_{ij} \in \mathbb{R}^{m \times n} \quad (226)$$

Furthermore, the sufficient condition for this linear transformation A to exist is given by the following theorem:

Theorem 5.1

If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, then $Df(a)$ exists iff all $D_j f^i(x_0)$ exist in an open set containing x_0 and if each function $D_j f^i$ is continuous at x_0 . Rmk. we call such f **continuously differentiable**

Complex derivative, on the other hand, is defined as:

Definition 5.1

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a complex function. The **derivative** of f at z_0 is defined as

$$\frac{df}{dz}(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \quad (227)$$

Or,

$$\frac{df}{dz}(z_0) = \lim_{\Delta z \rightarrow 0} \frac{u(z + \Delta z) - u(z)}{\Delta z} + \lim_{\Delta z \rightarrow 0} \frac{v(z + \Delta z) - v(z)}{\Delta z} i \quad (228)$$

where $u : \mathbb{C} \rightarrow \mathbb{R}$ is the real part of the complex function and v being the complex part.

This is *very different* from $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$! For the following two reasons:

Difference 1: Division

If we insist write the same definition for the real analog (i.e. denote a complex number $a + bi$ as $\begin{pmatrix} a \\ b \end{pmatrix}$, $f_1 \leftrightarrow u$, $f_2 \leftrightarrow v$), then the above definition would look like:

$$\lim_{(x,y)^T \rightarrow (x_0,y_0)^T} \begin{pmatrix} \frac{\partial f_1}{\partial(x,y)} \\ \frac{\partial f_2}{\partial(x,y)} \end{pmatrix} \quad (229)$$

Which doesn't make sense at all (Although one could argue $\frac{\partial f_i}{\partial(x,y)}$ is just $(\nabla f_i)^T$ but that's merely symbolic. What would division by a vector mean?).

Difference 2 : Mixing Of Components

To make it worse, even if we pretend such notation is sensible, they are still different. The limit of the real part $\lim_{\Delta z \rightarrow 0} \frac{u(z+\Delta z) - u(z)}{\Delta z}$ (which we hoped to correspond to $\frac{\partial f_1}{\partial(x,y)}$) has a complex part and similarly the limit of the complex part also has a real part! This means the components are naturally mixing.

So the real difference is that complex number should be viewed as a whole where division is well-defined. For me, if I ever found it hard to convince myself $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ is different from $\mathbb{C} \rightarrow \mathbb{C}$, I would try to answer the question “why can’t we use (a, b) to denote $a+bi$ ”? This tend to clear things up.

The next example illustrate this key difference:

Example 5.1 (A not differentiable complex function with its real analog differentiable)

Consider $f(x + iy) = x^2 + iy$. The real analog(Interactive Plot Of The Real Analog) is:

$$f \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x^2 \\ y \end{pmatrix} \quad (230)$$

This function is clearly differentiable since the jacobian is:

$$\begin{pmatrix} 2x & 0 \\ 0 & 1 \end{pmatrix} \quad (231)$$

which clearly continuous(component-wise) for all x, y .

However, this is not true for the complex analog. We now show the complex function is not differentiable at $0 \in \mathbb{C}$.

Suppose it is, then the two limits below should be equal at $0 \in \mathbb{C}$:

$$\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x + y_0 i) - f(x_0 + y_0 i)}{\Delta x} = \lim_{\Delta y \rightarrow 0} \frac{f(x_0 + (y_0 + \Delta y)i) - f(x_0 + y_0 i)}{i \Delta y} \quad (232)$$

But computing the limit tells us that:

$$2x_0 \neq 1 \quad (233)$$

Unless $x_0 = \frac{1}{2}$. Hence the function is only differentiable on the line $x_0 = \frac{1}{2}$.

One might ask, doesn’t the seem problem exists for the real analog? It’s quite obvious the answer is no, since we never compute a single limit of a finite difference.

In a way, $\mathbb{C} \rightarrow \mathbb{C}$ is like $\mathbb{R} \rightarrow \mathbb{R}$, we define the derivative to be the limit of finite difference. Yet the limits

$$\lim_{\Delta z \rightarrow 0} \frac{u(z + \Delta z) - u(z)}{\Delta z}, \lim_{\Delta z \rightarrow 0} \frac{v(z + \Delta z) - v(z)}{\Delta z} \quad (234)$$

resembles the $\mathbb{R}^2 \rightarrow \mathbb{R}$ where we expect the result to be independent of direction. That’s why the real analog is fundamentally different from the complex function itself.

Cauchy-Riemann Condition

Expanding (232), we would get:

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x + y_0 i) - f(x_0 + y_0 i)}{\Delta x} &= \lim_{\Delta y \rightarrow 0} \frac{f(x_0 + (y_0 + \Delta y)i) - f(x_0 + y_0 i)}{\Delta y} \\ \lim_{\Delta x \rightarrow 0} \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x} + i \frac{v(x_0 + \Delta x, y_0) - v(x_0, y_0)}{\Delta x} &= \lim_{\Delta y \rightarrow 0} \frac{u(x_0, y_0 + \Delta y) - u(x_0, y_0)}{i \Delta y} + i \frac{v(x_0, y_0 + \Delta y) - v(x_0, y_0)}{\Delta y} \\ \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} &= -\frac{\partial u}{\partial y} i + \frac{\partial v}{\partial y} \end{aligned} \quad (235)$$

Note that this equation essentially unwind the “mixing of components” we mentioned in Difference 2 : Mixing Of Components. Hence we get:

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}} \quad (236)$$

This is known as the Cauchy-Riemann Condition(C-R condition).

Let’s recall what had happened so far, we discussed the idea of complex derivative being a limit of finite difference leading to computing limits of the form $\mathbb{R}^2 \rightarrow \mathbb{R}$ (two input, one output) which puts additional restriction on differentiability. To be more specific, in order for these limit to exists, the value of the limit must be invariant of directions approaching the point of interest x_0, y_0 . Considering the specific direction along the real and complex axis we get the so-called CR condition.

We have effectively shown that CR condition is *necessary* for complex differentiability. It turns out however, that CR condition is also *sufficient*.

Theorem 5.2 (Cauchy-Riemann)

A complex function $f : \mathbb{C} \rightarrow \mathbb{C}$ is differentiable at $z_0 = x_0 + iy_0$ iff it satisfies the *Cauchy-Riemann condition*:

1. $u, v \in C^1$ (continously differentiable)
2. The partial derivatives satisfy

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \end{aligned} \quad (237)$$

In which case, the derivative is:

$$\begin{aligned} \frac{\partial f}{\partial z} &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \\ &= \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} \end{aligned} \quad (238)$$

Proof 5.1 (CR condition is sufficient)

We need to show, $\forall \epsilon > 0, \exists \delta > 0$ such that $|\Delta z| < \delta$ implies :

$$\left| \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} - \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) \right| < \epsilon \quad (239)$$

We consider the lineaer approximation of $f(z_0 + \Delta z) - f(z_0)$ around z_0 :

$$\begin{aligned} f(z_0 + \Delta z) - f(z_0) &= u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0) + i[v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0)] \\ &\quad + O(\Delta x^2) + iO(\Delta y^2) \\ &= \frac{\partial u}{\partial x} \Delta x + \frac{\partial u}{\partial y} \Delta y + i \left(\frac{\partial v}{\partial x} \Delta x + \frac{\partial v}{\partial y} \Delta y \right) \\ &\quad + O(\Delta x^2) + iO(\Delta y^2) \\ &= \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) \Delta x + \left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) \Delta y \\ &\quad + O(\Delta x^2) + iO(\Delta y^2) \end{aligned} \quad (240)$$

We need to expose $i\Delta y$, hence we rewrite the third term in the last equation above:

$$\left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) \Delta y \longrightarrow \left(-\frac{\partial u}{\partial y} i + \frac{\partial v}{\partial y} \right) i \Delta y \quad (241)$$

Therefore:

$$f(z_0 + \Delta z) - f(z_0) = \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) \Delta x + \left(-\frac{\partial u}{\partial y} i + \frac{\partial v}{\partial y} \right) i \Delta y + O(\Delta x^2) + iO(\Delta y^2) \quad (242)$$

Now applying CR condition:

$$\begin{aligned} f(z_0 + \Delta z) - f(z_0) &= \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) \Delta x + \left(\frac{\partial v}{\partial x} i + \frac{\partial u}{\partial x} \right) i \Delta y \\ &\quad + O(\Delta x^2) + iO(\Delta y^2) \\ &= \frac{\partial u}{\partial x} (\Delta x + i \Delta y) + i \frac{\partial v}{\partial x} (\Delta x + i \Delta y) \\ &\quad + O(\Delta x^2) + iO(\Delta y^2) \\ &= \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) \underbrace{(\Delta x + i \Delta y)}_{\Delta z} \\ &\quad + \underbrace{O(\Delta x^2) + iO(\Delta y^2)}_{O(|\Delta z| \Delta z)} \end{aligned} \quad (243)$$

Therefore the quotient is:

$$\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} + O(|\Delta z|) \quad (244)$$

and thus the quantity we want to control is:

$$\left| \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} - \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) \right| = O(|\Delta z|) \quad (245)$$

By definition of big-O, $O(|\Delta z|) < C|\Delta z|$ for $|\Delta z| \in \mathbb{R}$. Hence $\forall \epsilon > 0$, we can pick $\delta = \frac{\epsilon}{C}$ such that $O(|\Delta z|) < C\delta = \epsilon$

Analytical Function

Now we are ready for the definition of **analytic function**.

Definition 5.2 (Analytic Function on Entire Function)

- A complex function $f : \mathbb{C} \rightarrow \mathbb{C}$ is **analytic at** z_0 if it is differentiable at z and a neighborhood of z .
- A function that is differentiable throughout some open set $\Omega \in \mathbb{C}$ is **analytic in** Ω
- A function that is analytical throughout $\mathbb{C}$ is called **entire**.

Analytic is a much more restrictive property than merely differentiable (since it requires differentiability in the neighborhood). For instance, we can easily construct complex functions that are differentiable on *infinitely many points* in $\mathbb{C}$ but analytic *nowhere*.

Example 5.2 (Differentiable on infinite many points but analytic nowhere)

Suppose we want the function to be differentiable on $y = x$ and $y = -x$. Then we can, by CR, let:

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 2y \quad (246)$$

and

$$\frac{\partial v}{\partial x} = -2x, \quad \frac{\partial u}{\partial y} = 2y \quad (247)$$

A easy choice would just be:

$$\begin{aligned} f &= x^2 + y^2 + i(y^2 - x^2) \\ &= zz^* + i\left(\frac{z^2 + z^{*2}}{2}\right) \end{aligned} \tag{248}$$

Quick Exercise

Do you see why the above function is analytical nowhere?

Solution to 5.1

Every open set in $\mathbb{C}$ contains *some* point not on $y = x$ or $y = -x$

5.2 Complex Integrals

In lieu with the flow field interpretation of complex functions, generalizing the line integral for complex functions is very natural:

Definition 5.3 (Complex Integral)

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a continuous complex function and Γ is a smooth path in $\mathbb{C}$, then the limit on the R.H.S exists and we define the L.H.S integral by the limit.

$$\int_{\Gamma} f(z)dz := \lim_{\Delta z_i \rightarrow 0} \sum f(\xi_i)(z_{i+1} - z_i) \quad (249)$$

where $\xi_i \in (z_i, z_{i+1})$

We can compute the integral by computing two standard real line integrals:

$$\begin{aligned} \int_{\Gamma} f(z)dz &= \int_{\Gamma} (u + iv)(dx + idy) \\ &= \int_{\Gamma} (udx - vdy) + i \int_{\Gamma} (vdx + udy) \end{aligned} \quad (250)$$

We can define $A = (u, -v)$ and $B = (v, u)$ and $r = (dx, dy)$:

$$\int_{\Gamma} f(z)dz = \int_{\Gamma} A \cdot dr + i \int_{\Gamma} B \cdot dr \quad (251)$$

CHAPTER VI

Miscellaneous

6.1 Basic Numerics

Least Square

Suppose we have a dataset $\mathcal{D} = [(x_1, y_1), \dots, (x_n, y_n)]$, and by some prior knowledge we know there exists some relationship $y = kx + b$ in the dataset, the question is, how to find this particular relationship?

In otherword, we would like (k, b) to be able to predict y based on x that best adhere to the relationship implied by the dataset $\mathcal{D}$. The L_2 error of this predictor with respect to $\mathcal{D}$ is:

$$\mathcal{J} = \sum_i (y_i - (kx_i + b))^2 \quad (252)$$

The problem is now to find $\arg \min_{k,b} \mathcal{J}$. We know that we can reformulate this in matrix language:

$$\begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} b \\ k \end{pmatrix} \cong \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \quad (253)$$

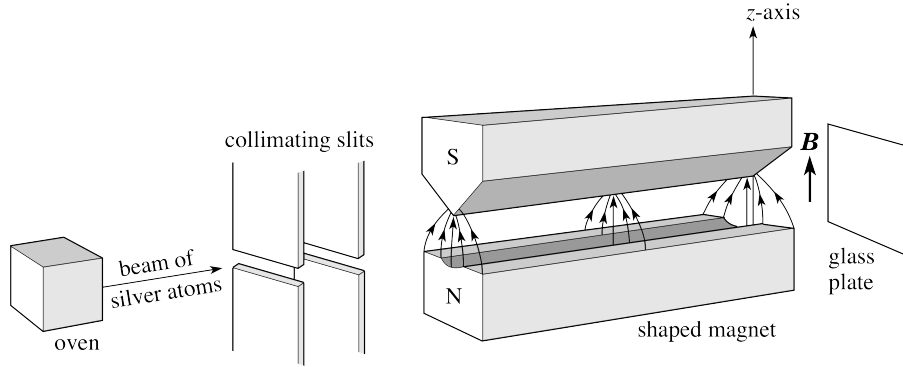
6.2 Classical Experiments in Physics

Stern-Gerlach Experiment

Slogan:

Stern-Gerlach demonstrated that the spin angular momentum of electron is quantized, with numerical values $\pm\hbar/2$

Consider the following experimental set up:



We want to know where would the silver atoms land on the glass plate. First we analyze through classical electromagnetics:

Theorem 6.1

The potential energy of interaction between a **magnetic dipole**(μ) and the **magnetic field**($\mathbf{B}$) is :

$$U = -\mu \cdot \mathbf{B} \quad (254)$$

Since the energy is lowered if the dipole aligns with the magnetic field. As a result, the force experienced by the the dipole is simply:

$$\nabla U = -\nabla(\mu \cdot \mathbf{B}) \quad (255)$$

In particular, we get that the z component of the force is :

$$F^{(z)} = -\partial_z(\mu^{(z)} B^x + \mu^{(x)} B^z + \mu^{(y)} B^y) \quad (256)$$

In our case,